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Clinical and statistical significance

1. A 44-year-old pharmacologist comes to the clinic for a routine follow-up of his hypertension. Current medications include lisinopril and atenolol. The physician tells the patient about a novel agent for hypertension that has recently been approved. A randomized-controlled trial found that use of this agent was associated with a 1 mmHg reduction in systolic blood pressure compared to lisinopril ($p = 0.04$). The patient asks the physician why bother mentioning the drug for his care if the reduction in blood pressure found in the study was so insignificant. Which of the following is the most appropriate response by the physician?
 - A) "Even though seemingly marginal, there was significance in this study."
 - B) "I'd like to discuss with you all potential options for your care."
 - C) "I think this agent could be better for you."
 - D) "The novel drug is significantly better so should replace lisinopril in your regimen."
 - E) "While clinically significant, the difference is not statistically significant as you've noted."

The correct answer is B.

The novel agent demonstrated a statistically significant reduction of blood pressure, albeit a marginal clinical significance – i.e., even if the agent reduces blood pressure 1 mmHg compared to lisinopril, the practical implications of this are likely ostensible at best. The patient, who is a pharmacologist, is astute to the concept of clinical trials and had asked a question to the physician that must be responded to openly and non-defensively.

Choice A ignores the patient's question and reiterates, "This drug had statistical significance albeit minimal clinical significance."

Choice C is a prevarication. If the drug likely demonstrates lack of clinical significance, why would the physician truly believe it would be better for the patient.

Choice D is evasive and imposing.

Choice E is inverted. The drug demonstrated statistical, not clinical, significance.

Public health screening

2. A neonate is about to receive a heel-prick test by an obstetrics nurse. The mother says she's okay with him receiving a vitamin K injection and hepatitis B vaccination, but doesn't want unnecessary interventions for her child. She says, "The notion of taking his blood for the sake of it? Don't you think we've already been through enough here?" Which of the following is the most appropriate response?
- A) "Detecting preventable causes of mental retardation as early as possible is important."
 - B) "High specificity of heel-prick testing as a screening implementation is advantageous in most scenarios."
 - C) "I understand your concerns. You can refuse it if you'd like. This is important to screen your baby for different conditions."
 - D) "Mam, this is not unnecessary. We need to screen your child for preventable causes of mental retardation."
 - E) "You've been through a lot. I understand."

The correct answer is C.

Heel-prick tests are done at birth to screen for conditions that, if not found as early as possible, could result in mental retardation for the child. PKU, galactose disorders, and congenital hypothyroidism are among the most frequently assessed conditions via heel-prick test.

Choice C communicates to the mom that she has the right to refuse the heel-prick test for her child, while simultaneously communicating that it is important to perform. If the mom refuses, this should be documented.

Choices A and B ignore the mother's concerns and merely reiterate the purpose of the test. Additionally, choice B is overly technical and verbose as a communication style in this setting.

Choice D merely refutes the mom's concerns without actually addressing them from a place of empathy.

Choice E is a nominal empathetic response in this scenario and does not communicate to the mom the importance of the test.

3. A 64-year-old man with a 35-pack-year history of smoking inquires about his risk for lung cancer. He says his wife keeps nagging him to ask the doctor about his risk of cancer. The physician tells him that, yes, he does fit the criteria in which a low-dose annual chest CT would benefit him for screening purposes. The patient replies, "You think I'm a moron. You wanna give me more radiation when that's not gonna do garbage for me at this point." Which of the following is the most appropriate response by the physician?
- A) "The best way to prevent lung cancer is to quit smoking. Would you like to discuss some possible methods for quitting?"
 - B) "Yes, it's radiation. A yearly x-ray is also an alternative for you if you don't want CTs."
 - C) "Yes, it's radiation. But a CT of your chest performed yearly is considered to be beneficial for you."
 - D) "You don't have to undergo CTs if you don't want to."

The correct answer is C.

The US Preventive Service Task Force (USPSTF) recommends annual low-dose CT scans of the chest for patients age 50-80, with at least a 25-pack-year history of smoking, and who currently smoke or quit within the past 15 years.

Choice C addresses the patient's concern about radiation while simultaneously upholding beneficence by communicating that CTs are still beneficial for him.

Choice B is wrong because x-rays are not an equivalent alternative to low-dose annual CT screening.

Choice D, although true, does not uphold beneficence. The physician still needs to communicate that the patient will benefit by receiving a low-dose annual chest CT. The patient can refuse, but this should still be communicated.

Choice A is a mechanical and sterile response that doesn't address the patient's concerns.

General patient communication

4. A 72-year-old woman with osteoporosis asks about any drug trials being conducted for her condition. The physician is aware of a recent drug that has been released onto the market, although he does not believe it should be used specifically in her case. He communicates this to the patient. She wants to start the drug anyway and says she will consider going to another physician who will be more lenient with her. What is the most appropriate response in this case?
- A) "I understand you want the best treatment possible, however this drug isn't ideal for you."
 - B) "I understand you want to try the new agent."
 - C) "I will transfer you to another physician but will first need to document this conversation."
 - D) "We should implement weight-bearing exercise for you."
 - E) "Why do you believe this drug would be helpful for you?"

The correct answer is A.

This response communicates both empathy for the patient's desire for optimal care while also being firm that the new agent is not ideal for her.

Choice B communicates empathy without conveying a clear stance on the novel agent, which the physician clearly believes is not suitable for the patient.

Choice C is an insensitive and tone-deaf response.

Choice D is first-line prevention and treatment for osteoporosis but doesn't relate to this scenario.

Choice E is wrong because the patient doesn't know anything about the new drug.

Bradford-Hill criteria

5. A clinical trial is performed to establish the relationship between a risk factor and a disease. Which of the following are among the criteria for establishing causality?
- A) Biological gradient, consistency of findings, sensitivity of test
 - B) Biological gradient, sensitivity of test, specificity of the association
 - C) Biologic plausibility, sensitivity of test, specificity of the association
 - D) Consistency of findings, specificity of the association, temporal sequence of association
 - E) Sensitivity of test, biologic plausibility, temporal sequence of association

The correct answer is D.

Student says, “Wtf?”

Students preparing for Step 3 should know the Bradford-Hill criteria, as these (annoyingly) show up on the MCQ component. Do they sound low-yield and dry? Yes. But it’s not our opinion that matters. It’s what shows up on the actual exam that does.

Students studying for Steps 1 and 2CK might benefit from being peripherally aware of the criteria, since recent changes in the USMLE might entail these worthwhile to know.

Choice D is the only one that does not include sensitivity of test, which does not help determine causation.

The Bradford-Hill criteria for establishing causation in epidemiology:

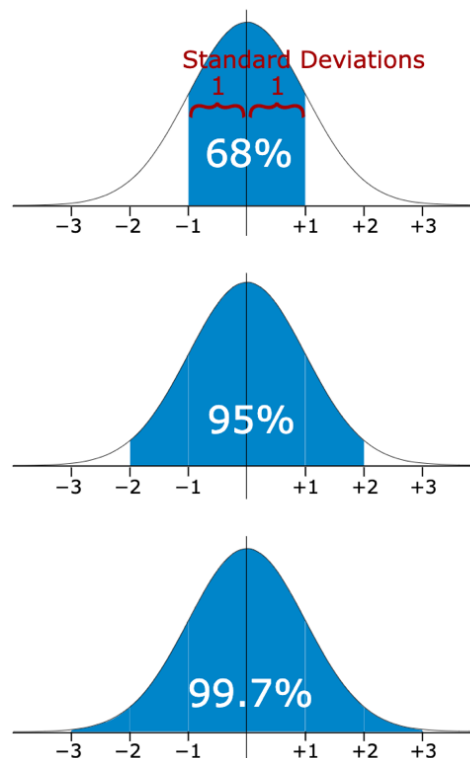
- Strength (effect size)
- Consistency (reproducibility)
- Specificity (no other likely explanation)
- Temporality (cause precedes the effect)
- Biological gradient (dose-response relationship)
- Biologic plausibility (plausible mechanism)
- Coherence (between epidemiological and laboratory findings)
- Experiment (experimental evidence)

Data interpretation

6. A 62-year-old man is enrolled in a clinical study looking at an enzyme marker for pancreatic cancer. The enzyme levels in the population are normally distributed, with an average of 100 mg/dL and standard deviation of 10 mg/dL. Increased serum levels are directly correlated with increased risk of cancer. It is determined that 2-3% of people would benefit from an annual CT of the abdomen. The patient's serum levels are 110 mg/dL. The patient asks if he should receive CTs but doesn't want radiation. Which of the following is the most appropriate response to this patient?
- A) "I understand medical tests are annoying. We don't have to do things you don't want."
 - B) "I understand you don't want to receive radiation and that's okay."
 - C) "The results suggest you wouldn't benefit from CTs."
 - D) "Would you mind please telling me more about your understanding of CT scans?"
 - E) "You don't need to have any radiation if you don't want, but the results suggest you'd benefit."

The correct answer is C.

The patient wouldn't benefit from CTs, so this should be communicated to him plainly.



If 2-3% of people would benefit from a CT scan, then those who have serum levels of the marker greater than ~2 standard deviations above the mean should be considered for this diagnostic intervention.

Roughly 95% of the population falls within 2 standard deviations +/- of the mean, which means roughly 2.5% fall below 2SD, and 2.5% above 2SD.

120 mg/dL = 2 standard deviations above the mean = ~97.5%tile → roughly 2.5% of the population has serum levels above 120.

80 mg/dL = 2 standard deviations below the mean = ~2.5%tile → roughly 2.5% of the population has serum levels below 80.

80-120 mg/dL = +/- 2 SD from the mean = ~95% of the population has serum levels within this range.

Since we are only concerned with patients who have high serum values, the top 2-3% (i.e., those who are >2SD above the mean) should be screened with CT.

This patient has a result of 110 mg/dL, indicating he is only 1 SD above the mean (~84%tile). +/-1 SD from the mean is +/- 34% of the population, where 68% of people fall between +/-1SD (in this case, 90-110 mg/dL).

Clinical trials

7. A pharmaceutical company is working on developing a new drug for heart failure. They believe they have discovered a compound, sacubitril, which when combined with the frequently prescribed valsartan, could be used as a combination therapy for heart failure. Which of the following study designs would be most appropriate for phase III?
- A) Administration of sacubitril to a small number of healthy volunteers to assess safety
 - B) RCT comparing the efficacy of valsartan with the valsartan/sacubitril combination
 - C) RCT comparing the efficacy of valsartan versus sacubitril alone
 - D) RCT comparing the efficacy of valsartan/sacubitril combination versus valsartan/placebo
 - E) Releasing sacubitril to the market and allowing physicians to report side effects

The correct answer is D.

Phase	Purpose and design
0	- Very small # of volunteers - Pharmacokinetics + pharmacodynamics - Determines if drug interacts with its expected target - Single, low dose (<1% of calculated dose required to produce clinical effect) - Not expected to show clinical effect or significant adverse effects - Helps choose between chemical analogues for further investigation
I	- “Is it safe?” - Small # of volunteers - Dosage range for safety - Escalation of dose ends when unacceptable side-effects occur; prior dose is determined to be the maximum dose tolerated - Identifies common side-effects
II	- “Does it work?” - Small-moderate # of patients with disease - Explores efficacy within safety range - Multiple doses to establish efficacy, often against placebo - Reveals less common side-effects
III	- “Is it better?” - Larger number of patients randomly assigned to the study drug or standard of care - Final determination/confirmation of efficacy and safety - Multiple doses confirming clinical efficacy - Compares study drug to current standard of care
IV	- “What type of efficacy and adverse effects am I observing in my personal practice?” - Post-marketing surveillance of patients following approval of the drug - Healthcare practitioners report any adverse effects encountered in clinical practice - Provides surveillance for rare outcomes

Phase III is characterized by comparing the new drug, sacubitril, with the current treatment, valsartan, alone or in combination. Both answers C and D refer to phase III studies, however the study hypothesis is that efficacy will be demonstrated specifically when sacubitril is combined with valsartan. The aim of the study is not to demonstrate efficacy of sacubitril alone versus valsartan alone.

It should be noted that valsartan/sacubitril is a real drug combination sometimes used for heart failure. Sacubitril is a neprilysin inhibitor. Neprilysin is an enzyme that breaks down atrial natriuretic peptide. Valsartan is an angiotensin II receptor blocker.

Phase 0 would refer to testing the pharmacokinetics and -dynamics sacubitril on a very small group of healthy volunteers and/or patients with disease.

Phase I would entail testing for the safety, toxicity, and dosage of sacubitril.

Phase II testing would assess for efficacy of the drug (i.e., for effects on BP, EF, etc.).

Phase IV is the answer if physicians report on their findings and/or adverse effects of the drug after it has already been on the market – i.e., post-marketing surveillance.

8. A 45-year-old woman with advanced breast cancer is considered for a clinical trial of a novel chemotherapeutic agent. She would be one of 50 women with advanced breast cancer receiving this drug. Physicians would monitor the clinical response to different doses. Which clinical trial phase would this patient be a part of?
- A) Phase 0
 - B) Phase I
 - C) Phase II
 - D) Phase III
 - E) Phase IV

The correct answer is C.

Phase	Purpose and design
0	- Very small # of volunteers - Pharmacokinetics + pharmacodynamics - Determines if drug interacts with its expected target - Single, low dose (<1% of calculated dose required to produce clinical effect) - Not expected to show clinical effect or significant adverse effects - Helps choose between chemical analogues for further investigation
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Phase II clinical trials entail a small to moderate number of patients with disease who are administered multiple doses of a drug to assess for clinical efficacy.

9. A pharmaceutical company has developed a new immunotherapy agent, Drug X. Early studies demonstrated preliminary efficacy against both metastatic melanoma and actinic keratosis. However, studies also found that 10% of patients developed severe myelosuppression and 2% died. Which of the following is the most likely decision to be made by the FDA regarding this agent?
- A) Another study will be requested
 - B) The novel drug will be approved for both conditions
 - C) The novel drug will be approved for metastatic melanoma only
 - D) The novel drug will be approved for actinic keratosis only
 - E) The novel drug will not be approved for either condition

The correct answer is C.

Risk/benefit analysis is an important step in determining whether a novel treatment should be used in clinical practice. The benefits of treatment need to be weighed against risks. In this case, we have a novel agent that causes severe myelosuppression in 10% and fatality in 2%, making it very dangerous.

Although a precursor to SCC, actinic keratosis carries excellent prognosis and is curative with treatment methods such as cryotherapy. Using a novel therapy with high morbidity and mortality would be inappropriate.

On the other hand, metastatic melanoma carries a very low survival rate and effective management options are limited. Therefore, the benefits of this drug make FDA approval for only this condition most likely.

Children enrolled in research

10. A 14-year-old girl comes to the physician with her mom for a routine health maintenance examination. The physician mentions a new study that would require the patient's data be collected for the purposes of research. Consent is required by whom in this case?
- A) Both parents
 - B) One parent only
 - C) Patient only
 - D) Patient and one parent
 - E) Patient and both parents

The correct answer is E.

This is a “weird” question but valuable for USMLE.

US federal requirements for children involved in research state that **both parents** must give consent when at all possible, in addition to the patient who is the minor.

One parent may give consent if the second parent is unavailable or deceased.

It is never appropriate to enroll a minor in a research study without parental consent.

Choice of study type

11. A group of researchers is interested in finding out whether IV drug-use by patients at a select hospital is associated with increased risk of atrial myxoma compared to non-drug-users. A study is performed by searching through medical records of patients who have history of IV drug-use versus the medical records of those who do not have history of IV drug-use. Which of the following best describes this study design and why it was chosen?
- A) Case-control study; myxoma cases not well-documented at the select hospital
 - B) Case-control study; myxoma cases well-documented at the select hospital
 - C) Prospective cohort study; myxoma cases not well-documented at the select hospital
 - D) Prospective cohort study; myxoma cases well-documented at the select hospital
 - E) Retrospective cohort study; myxoma cases not well-documented at the select hospital
 - F) Retrospective cohort study; myxoma cases well-documented at the select hospital

The correct answer is E.

Cohort studies always move in the direction of risk factor → disease. That is:

Group with risk factor vs Group without risk factor → did they develop the disease.

Case-control studies are the opposite. They move in the direction of disease → risk factor. That is:

Group with disease vs Group without disease → did they have the risk factor.

Case-control studies are ideal for assessing rare conditions. If you want to determine whether, e.g., a rare type of pancreatic cancer is caused by smoking, it is not feasible to do a cohort study because this would entail needing to follow smokers vs non-smokers and seeing whether over time they happen to specifically develop this rare type of cancer.

Instead, a case-control study would enable first starting with a group of patients who have the rare type of cancer and then simply looking back to see whether they were smokers or not.

So where does retrospective cohort study come into play?

Cohort studies are almost always prospective. Once again, they are always in the direction of risk factor → disease. *But if the process has already happened and you're looking at medical records,* then your observations are retrospective, and we call it a retrospective cohort study.

If a hospital has not specifically documented its myxoma cases, then there's no way to start with a case-control study because we don't have a way of assembling a case (disease) group. In this scenario we'd be forced to say, "Well our original question is whether IV drug-use relates to development of myxomas, but we don't have a way of isolating out those who developed myxomas, so we'd have to manually and clumsily look back through the medical records of IV drug-users vs non-drug users to see who developed myxoma at this hospital."

Bottom line: Case-control studies are always disease → risk factor. Cohort studies are always risk factor → disease. Almost all cohort studies are prospective. Retrospective cohort studies are applied in a public health setting when the research entails looking back through the medical records of those who had a risk factor vs no risk factor and seeing whether they developed the disease.

Angry patients

12. A 25-year-old woman presents to the physician for a follow-up appointment. The physician arrives to see the patient ten minutes after the scheduled time. The patient is very angry. She says to the physician, "You strolling in here all pompous like that. You guys think it's okay to just keep people waiting because you're doctors." Which of the following is the most appropriate response to this patient?
- A) "Did something happen on the way to the clinic that has made you upset?"
 - B) "I'm really sorry, I was caught up in a meeting. How are you doing today?"
 - C) "I'm sorry to keep you waiting. Let's discuss why you've come in today."
 - D) "I think it's best that we reschedule."
 - E) "I understand you are upset I am late, but it's not appropriate to raise your voice here."

The correct answer is C.

If a patient is angry that you are late, a quick apology, followed by immediately addressing why the patient has come to see you, is the most appropriate response.

Choice B is wrong because you do not need to justify or explain to a patient why you may have been late. A quick apology, followed by immediately addressing why the patient has come to see you, is most appropriate.

Choice A is presumptuous. Any concerns the patient has should be addressed during the meeting, but the best initial response to the angry patient is a quick apology, followed by moving the consult forward.

Choice D is inappropriate since the patient should not be dismissed simply because she is upset.

Choice E is patronizing.

In-hospital risk management

13. A community hospital has experienced three cases of wrong-limb surgery during the past five years, where the wrong limb of a patient was operated on. The hospital is now legally required to implement policies to prevent recurrence. One of the measures involves writing an X on the appropriate limb to be operated on. It is most appropriate for the X to be written by whom?
- A) Anesthesiologist under the supervision of an awake patient
 - B) Anesthesiologist while the patient is asleep
 - C) Operating surgeon
 - D) Operating surgeon under supervision by anesthesiologist
 - E) Patient

The correct answer is C.

The individual who is performing the surgery draws the X. The attending surgeon does not need to be supervised doing this.

The patient is not the one who draws the X. A student might say, “Well how is it possible that the patient him or herself literally wouldn’t know which limb is to be operated on?” But the rule is: he/she who is performing the surgery is the one who draws the X.

Bottom line: The USMLE asks this question on wrong-limb surgery. The question might be formatted in an EMB/ethics-type of vignette.

Prisoners as participants in research

14. A new insulin pump is being investigated for the management of patients with diabetes mellitus. Among the populations being studied is that of a local prison. The details of the study proposal are being evaluated. Which of the following statements is true regarding prisoner participation in research?
- A) All research involving prison inmates is ethically problematic
 - B) Compensation prisoners receive can be reduced by adjusting for costs-of-living intra-prison
 - C) Early release is an acceptable form of compensation for participating in a study
 - D) Prisoners can participate under the exact same conditions as non-prisoners
 - E) The inmates cannot refuse to participate in clinical research trials

The correct answer is D.

Prisoners can participate under the same conditions as non-prisoner groups. However, coercion is a significant risk when it comes to any vulnerable group, including prisoners, and all care must be taken to ensure participation is out of one's own volition.

Informed consent must always be obtained irrespective of the nature of the group of participants.

Compensation for prisoners must not differ from that of the other participants, and they cannot be offered early release in exchange for partaking in the study.

Confidentiality for minors

15. A 14-year-old girl presents to the clinic with her mom for burning pain on urination. A urinalysis is performed and confirms the diagnosis of a urinary tract infection. The physician speaks to the patient in private and the girl confides that she has been sexually active. She says she does not want the physician to tell her mom about either her UTI or her sexual activity. What is the most appropriate response by the physician?
- A) "Don't worry, I won't talk to her about anything."
 - B) "I understand, but I am legally required to discuss these things with a parent."
 - C) "I will not talk to her about your sexual activity, but I must discuss the UTI with her."
 - D) "If I don't mention your UTI to her, I won't be able to give you antibiotics, but you need them."
 - E) "Would you mind telling me why you don't want me to discuss these things with her?"

The correct answer is C.

Minors have the right to confidentiality *without* parental consent for sexual health, obstetric/gynecologic care, drug-use, and/or mental health management.

In this case, the minor has a right to have her sexual activity withheld from her parents. Breaching this confidentiality jeopardizes non-maleficence – i.e., it does harm if she believes there will be punitive measures against her if her mom knows. The result is she may not seek continued care with the physician, or may learn to distrust doctors in general, putting her at greater risk for problems going forward.

Even though sexual activity is one of the greatest risk factors for UTIs, the latter in and of themselves do not fall under the envelope of conditions that can be withheld from parents. This child needs antibiotics for her UTI. The consent of her mom is required. This needs to be communicated to the minor.

If the minor is emancipated, then no parental consent is required, even for UTI management.

Emancipated minors are those who are not dependent on their parents for everyday life and are therefore eligible to make healthcare decisions without parental consent. This refers to minors who are in the military, living independently (e.g., a high-school graduate with a full-time job who lives away from home), or who have children of their own whom they support. Non-emancipated minors require parental consent for most conditions until the age of 18. An important exception is healthcare in a life-threatening situation (e.g., emergency blood transfusions).

16. A 16-year-old girl presents to the clinic with her mother for a routine physical. The mother asks her daughter during the consult if she has been sexually active and using marijuana. The patient does not immediately respond. The physician kindly asks for the mother to step out so that he can speak to the patient alone. The mother refuses. Which of the following is the appropriate response by the physician?
- A) "I will kindly ask you to step out."
 - B) "It is important that I speak to your daughter in private."
 - C) "No problem."
 - D) "That is okay, but I will need to document that we have not been able to address all aspects of your daughter's health."
 - E) "Would you like for one of our nurses to sit in with the two of us?"

The correct answer is E.

Offering to have a female member of staff sit in during the consult is one effective way to make parents feel more comfortable leaving their children alone with a physician.

If the mother ultimately refuses to allow her daughter be left alone with the physician, this must be respected, but also documented. However offering to have a female member of staff sit in during the consult is a solution that should first be attempted.

If the physician suspects child abuse, he or she must contact child protective services, but this is not advised in a setting where abuse is not specifically suspected.

Choice A is dismissive of the parent's refusal.

Choice B is a reiteration of what the physician already attempted prior to the parent's refusal.

17. A 15-year-old female presents to the clinic accompanied by her mother for a routine health maintenance examination. The physician would like to ask more details about her sexual history. Which of the following is the most appropriate next step in communication?
- A) Ask questions while both the patient and mother are present together
 - B) Ask the mother to leave the room before asking the patient questions
 - C) Ask the patient to leave the room before asking the mother questions
 - D) Ask the patient in front of her mother if she is okay answering questions about sexual history
 - E) Have the mother step out of the room and call for a female nurse to be present

The correct answer is B.

Similar to the previous question, the first step in attempting to ask a minor questions is kindly asking the parent to step out. If the parent refuses, offering for a female member of staff to sit in is sometimes effective. If the parent refuses again, his or her wishes must be respected, followed by proper documentation by the physician. If the physician does not suspect abuse, there is no need for reporting. Parents have a right to refuse physicians spending time with their children.

Hospice care

18. An 81-year-old man has recently been diagnosed with stage 2 pancreatic cancer and malignant pleural effusion. The patient is not interested in aggressive treatments and would like to be referred for hospice care. The physician explains that without treatment his life expectancy would be approximately 8 months. Which of the following would make this patient ineligible for hospice referral at this time?
- A) Insufficient stage of cancer
 - B) Patient prognosis
 - C) Patient declination of aggressive treatment
 - D) Periodic thoracentesis
 - E) There are no contraindications for hospice care at this point in time

The correct answer is B.

Eligibility for hospice care necessitates terminal illness with a prognosis of <6 months of life remaining. The patient must also not be pursuing, or interested in pursuing, any type of life-extending management.

The stage of pancreatic cancer is irrelevant. If the patient had Stage 1 cancer and a 4-month prognosis, he would be eligible for hospice care insofar as he is not interested in pursuing life-extending measures.

Types of management that make a patient ineligible for hospice care include any type of chemo- or radiotherapy, as these extend life.

Sounds highly specific and weird, but USMLE 2CK wants you to be aware that thoracentesis *is an acceptable* management to be received by those in hospice care. The implication is that those with, e.g., malignant pleural effusions, may receive thoracentesis to improve symptoms, but the management in and of itself does not extend life.

19. A 42-year-old man with terminal amyotrophic lateral sclerosis (ALS) was transferred to hospice care one week ago. The patient says he would like to take part in a clinical trial of a new drug for advanced ALS that may improve prognosis. Which of the following is the most appropriate next step in management?
- A) Allow the patient to leave hospice and inform him he will not be eligible to return
 - B) Allow the patient to partake in the clinical trial while in hospice
 - C) Do not allow the patient to leave the hospice
 - D) Request psychiatric evaluation of the patient
 - E) Transfer the patient to palliative care

The correct answer is E.

Patients who are receiving hospice care are free to leave at any point if they wish to pursue more aggressive management of their condition. The requirements for hospice admission are terminal illness with prognosis of < 6 months and no pursuit of life-extending treatments.

Choice A is wrong because the patient is free to resume hospice care if he wishes to terminate his prescription of the new drug. While patients are allowed to take medication while in hospice, this new drug may improve prognosis and therefore is considered life-extending, meaning it cannot be taken while in hospice care.

The aims of palliative care are to optimize quality of life and mitigate suffering in patients who may have >6 months of life remaining and who may be seeking life-extending management.

Palliative care happens in hospital. Hospice care occurs wherever the patient calls home.

Palliative care may be paid by insurance or self-pay. Hospice care is covered by Medicaid, Medicare, or insurance.

Bottom line: Patients in hospice who wish to pursue life-extending management may be transferred to palliative care.

Futile treatment

20. A 75-year-old man presents to the emergency department with sudden-onset severe headache, nausea, vomiting, photophobia, and neck stiffness. He was in his usual state of health until this morning when the symptoms began. His past medical history is significant for hypertension managed with hydrochlorothiazide and lisinopril. He was in a serious car accident 20 years ago that left him in a coma for one month. A non-contrast CT of the head confirms subarachnoid hemorrhage. Despite the best efforts of the physicians, the patient's condition deteriorates and he becomes non-responsive over the next few hours. Brain-stem reflexes are absent. The patient cannot breathe spontaneously and is pronounced brain-dead. The physicians are preparing to discontinue life support. However, the patient's spouse and children insist that the patient had beaten the odds multiple times before and that they are certain he will recover again. Moreover, they have seen many reports about miraculous recoveries. They threaten to sue the physician if life support is discontinued. Which of the following is the most appropriate action by the physician?
- A) Continue life support
 - B) Consult the hospital ethics committee
 - C) Do an electroencephalogram (EEG)
 - D) Transfer care to another physician
 - E) Withdraw life support

The correct answer is E.

Physicians are not required to provide futile care when there is no prospect of the patient recovering. In this case, the patient is brain-dead and further treatment would not change his prognosis.

EEG and MR angiography can be considered in uncertain cases but are not required to diagnose brain-death in this patient.

Absence of brainstem reflexes and apnea test prove brain-death. Coma and brain-death should not be confused. Treating a brain-dead patient is futile, while treating a comatose patient is not.

The hospital ethics committee can be consulted when there are differences in opinion between family members regarding what an incapacitated patient might want in terms of care.

Transfer of care to another physician is the answer on the USMLE if a patient wishes to discontinue life support but the physician wishes not to do it. This is on the NBME exams. The implication is that care will be discontinued as per the patient's autonomy and wishes, but the physician who is strongly opposed to the discontinuation distances him or herself from the decision for emotional reasons. Transfer of care to another physician is acceptable in this scenario.

Transfusion in Jehovah's witnesses

21. A 35-year-old Jehovah's witness and her 16-year-old daughter are brought to the emergency department by the paramedics following a serious car accident where they lost a lot of blood. The patients are being prepared for surgery and the mother insists that neither her nor her daughter be given any blood products, regardless of the circumstances. The daughter is conscious and nods her head in agreement. Soon afterwards, both patients lose consciousness and transfusions are deemed necessary. Which of the following is the appropriate course of action?
- A) Administer saline only to both patients
 - B) Administer a blood transfusion to both patients
 - C) Administer a blood transfusion to the daughter only
 - D) Seek a court order to give transfusion to both patients
 - E) Seek a court order to give transfusion to the daughter only

The correct answer is C.

Minors who are Jehovah's witness should be transfused in life-threatening emergencies, even if they are emancipated. Parents cannot refuse life-saving transfusions for their children.

If a blood transfusion is not immediately life-threatening (e.g., severe anemia), the parent has the right to refuse it for her child.

Involuntary hospitalization

22. A 45-year-old man with history of schizophrenia is brought to the emergency department after being seen dressed up as a cowboy on the street with a gun, claiming he needed to “kill the evil.” The patient appears disheveled and smells of alcohol. His thoughts are disorganized and he is unable to provide any meaningful history. He does not want to stay at the hospital. Fifteen minutes later, the patient’s wife shows up at the emergency department and profoundly apologizes for the patient’s behavior and insists that this will never happen again. She also wants the patient to be discharged from the hospital. Which of the following is the best course of action?
- A) Admit the patient involuntarily
 - B) Discharge the patient and schedule outpatient psychiatry follow-up
 - C) Discharge the patient and schedule regular visits from the social worker
 - D) Make both the patient and his wife sign the request to be discharge against medical advice
 - E) Make the patient sign the request to be discharged against medical advice

The correct answer is A.

While patients have the right to refuse medical care once they are fully aware of the potential implications of their decisions, they can nevertheless be hospitalized involuntarily if they are considered a danger to themselves or others. This patient is exhibiting signs of psychosis and has expressed homicidal ideation. He cannot be safely discharged from the hospital and needs to be admitted for observation in the emergency department with a psychiatric team called to evaluate.

Terminal diagnoses

23. A 75-year-old man with cholangiocarcinoma presents for an appointment to discuss his recent PET scan. New hepatic metastases are identified. The patient appears despondent and has little interest in pursuing further treatments. At the end of the visit he asks the physician to prescribe him enough opioids to “go to sleep and never wake up.” Which of the following is the most appropriate next step by the physician?
- A) Ask the patient why he made such a request
 - B) Consult psychiatry
 - C) Hospitalize the patient involuntarily
 - D) Hospitalize the patient voluntarily
 - E) Prescribe meperidine

The correct answer is A.

The patient has received difficult news and is at high risk of depression. He needs to be communicated with empathetically and provided all support services possible. Further discussion with the patient will be beneficial for his ability to cope with the poor prognostic information.

Psychiatric liaison may provide benefit for this patient if he develops major depression due to a medical disorder. However in this current scenario, the patient's thought process merely needs to be explored further. Consultation with psychiatry is premature.

Hospitalizing the patient, voluntarily or involuntarily, is not necessary at this time. If he communicates suicidal ideation, involuntary admission to hospital would be required.

Meperidine is a highly addictive opioid. Although the patient is visibly distressed about the news, prescription of opioids at this time would be unethical.

Pain management in terminally ill patients

24. A 60-year-old male with advanced pancreatic cancer has been transferred to hospice. He is suffering from severe pain and on a fentanyl patch. While crying, he says, "I just want all of this to end. I'm really unhappy. I just don't want to be depressed anymore." Which of the following is the most appropriate next best step by the physician?
- A) Administer additional pain medication at low doses
 - B) Administer high-dose opioids despite risking respiratory depression
 - C) Ask the patient what upsets him
 - D) Commence the patient on sertraline
 - E) Consult psychiatry

The correct answer is B.

Patients who are in severe pain can appear as though they have major depression even when they do not. Pain control is one of the most important facets of ethics for USMLE. Always fully treat pain, even if the patient has history of drug abuse.

It is acceptable to give high doses of pain medication that may inadvertently facilitate end of life. When you are answering tough questions on USMLE, the general rule is: never withhold adequate pain management no matter what. It is never okay to allow patients to suffer from pain.

It should be noted, in contrast, that physician-assisted suicide (e.g., prescribing high doses of benzodiazepines to a patient who wants to end his or her life) and euthanasia (intentionally ending a patient's life, unrelated to pain management) are illegal in most of the United States. Ethical views vary among physicians.

The USMLE will not ask questions with equivocal answers. Your knowledge going into the exam should be simply:

Physician-assisted suicide and euthanasia are not acceptable. Always fully treat pain.

25. A 78-year-old man with a past medical history significant for terminal multiple myeloma has been hospitalized due to hip fracture. The patient has been bed-bound for the past month and is deemed a poor surgical candidate. Morphine is used to manage pain. Following admission, the patient develops pneumonia and a decision is made to provide comfort care only. The patient's respiratory rate is 6/min with an oxygen saturation of 89% on room air. Blood pressure is 90/60 mmHg. He still complains of unbearable pain. Which of the following is the most appropriate next step in management?
- A) Administer naloxone
 - B) Decrease the dose of morphine and administer oral ibuprofen
 - C) Increase the dose of morphine
 - D) Maintain the dose of morphine and add intravenous acetaminophen
 - E) Send for a ventilator

The correct answer is C.

Increasing the dose of morphine increases the immediate risk of death for this patient. However this patient has already been deemed terminally ill and has been put on comfort measures only. While physician-assisted suicide (with the exception of a few states) is illegal, interventions that may hasten the end of life are considered ethically permissible if they aim to reduce pain.

It is the aim of the intervention that is important. This patient is still in pain. Increasing the dose of morphine is appropriate regardless of whether it may hasten end of life. Prescribing medications to a patient with the aim of facilitating death, on the other hand, is unethical, and almost always illegal.

Intravenous acetaminophen is highly effective for minor pain. In the setting of a patient on opioids with cancer pain who needs even more pain relief, acetaminophen is unlikely to be effective. The difficult takeaway here is: fully treat pain, even if it facilitates respiratory depression and end of life.

Bottom line: Managing pain is ethically permissible even if it hastens end of life.

Establishing competence

26. A 77-year-old man with Alzheimer disease is being evaluated for pneumonia. However, upon suggesting hospitalization and antibiotic management, the patient declines treatment. The neurologic examination shows poor short-term memory and no focal findings. He can follow simple commands and has intact retrograde memory. Which of the following is most accurate with respect to this patient?
- A) The diagnosis of Alzheimer disease makes the patient incompetent
 - B) The normal neurologic examination makes the patient competent
 - C) The patient's diminished short-term memory makes him incompetent
 - D) The patient's intact retrograde memory makes him competent
 - E) The patient's understanding of the consequences of his decision and the rationale provided make him competent

The correct answer is E.

Adult patients are presumed to be competent to make healthcare decisions until proven otherwise. It is very difficult to show an adult is incompetent. The most important aspect of determining competence is determining whether the patient understands the possible consequences of his or her decision and can provide some form of rationale for it, even if the physician disagrees with the patient's sentiments.

This patient states he does not wish to undergo any treatment. He is entitled to make such a decision. The diagnosis of Alzheimer disease does not make one incompetent, nor does impaired short-term memory.

Competence does not depend on the neurologic exam unless the patient is in a state of delirium that interferes with the ability to be examined.

Retrograde memory has no impact on one's competence to make decisions regarding personal care.

Decision-making for the incapacitated patient

27. A 78-year-old man experiences an acute ischemic stroke. Despite treatment, the patient's condition deteriorates. He is able to articulate that he would like all life-saving measures to be performed. He also has a living will expressing similar wishes. The patient then passes into a vegetative state. The patient is accompanied to the hospital by his spouse and two adult children. The spouse is the durable power of attorney and wishes to discontinue life-saving care. Which of the following is correct in terms of making health-care decisions for this patient?
- A) Follow the living will and verbal wishes of the patient
 - B) Follow the wishes of the spouse
 - C) The hospital legal team should be consulted before consulting with family
 - D) The hospital legal team should be consulted before following the living will
 - E) Two or more adult children present may override a living will

The correct answer is B.

A durable power of attorney in healthcare is a person designated to make decisions on behalf of a patient. If the patient becomes incapacitated, the durable power of attorney equals the patient for all intents and purposes, surpassing in priority any verbal wishes the patient may have made prior to incapacitation, as well as the patient's living will.

In other words, the durable power of attorney can make all personal and legal decisions on behalf of the patient if he or she becomes incapacitated.

There are four main types of Power of Attorney:

General power of attorney: the designated individual can perform most acts as the principal, including managing finances and opening accounts. The general power of attorney arrangement is terminated if the patient becomes incapacitated.

Durable power of attorney: the designated individual can act on the principal's behalf and includes a durable clause that maintains this power even if the patient becomes incapacitated.

Springing durable power of attorney: the designated individual becomes the durable power of attorney only when the patient becomes incapacitated. Prior to incapacitation, the designated individual does not act as a power of attorney for the patient.

Special or limited power of attorney: the designated individual has power limited to a specific area, such as selling the property of the patient.

Bottom line: A durable power of attorney overrides the living will (advanced directive) in the setting of patient incapacitation, even if the patient specified wishes to the contrary immediately prior to incapacitation.

Do not resuscitate

28. An 80-year-old woman is admitted to hospital for chest pain, dyspnea, and diaphoresis for the past three hours. The patient has a long history of cardiac disease and has previously signed a do-not-resuscitate (DNR) form. Subsequently, she is diagnosed with the acute ST-elevation myocardial infarction. Which of the following is the most appropriate treatment for this patient?
- A) Ask the patient if she wishes to override her DNR form
 - B) Immediately discharge the patient
 - C) Perform percutaneous coronary intervention
 - D) Provide comfort care only
 - E) Request the patient rescind the DNR form before treatment

The correct answer is C.

DNR stands for “do not resuscitate”. It does not stand for “do not provide adequate medical care.” If this patient were to develop cardiorespiratory arrest, it would be inappropriate to resuscitate her as this would clearly violate her directive. However, the DNR form does not apply to any other aspect of patient care and all other appropriate measures should be implemented as they would for any other patient. Therefore, discharging the patient or withholding care would be inappropriate.

Bottom line: DNR forms are a legal order whereby a patient may refuse cardiopulmonary resuscitation if his or her heart stops. This does not apply to interventions such as percutaneous coronary intervention in the treatment of acute myocardial infarction.

Patient privacy and medical records

29. A 65-year-old woman has been hospitalized due to an exacerbation of COPD and subsequent pneumonia. Her status has deteriorated significantly over the past year and this is her 6th hospitalization this year. The patient understands that her prognosis is poor but nevertheless wants to continue with aggressive treatments. She does not want any of her family members to know that she may not survive this hospitalization. Upon leaving the patient's room, the physician is confronted by the patient's husband who demands to know the patient's prognosis. He says that their children live out of state and need to be informed. He is her durable power of attorney. Which of the following would be the appropriate response by the physician?
- A) "Have you spoken to your wife about this?"
 - B) "I am sorry but cannot discuss her prognosis at this time."
 - C) "I am sorry but you will need to file an official request for release of medical records."
 - D) "I am sorry but your wife's condition is serious and the next few hours are critical to see how she responds to treatment."
 - E) "Your wife is stable and will make a full recovery."

The correct answer is B.

This is a “hard” question.

The patient clearly stated that she does not want her prognosis shared with any family members. This must be firmly respected by the physician. The patient’s spouse, although the durable power of attorney, only acts on behalf of the patient in the medical setting if she becomes incapacitated. While the patient is awake and communicating, the patient’s wishes must be followed.

Choice A is wrong because it equivocates the wishes of the patient despite the fact that they have already been firmly expressed by her. The physician needs to respect the wishes of the patient and communicate to the spouse that her prognosis cannot be discussed right now. If the physician responds with choice A, then this might put the patient in an uncomfortable position where she feels compelled to discuss her condition further despite not wanting to. Had the patient not explicitly communicated a preference, *then* choice A might be more appropriate.

Partner notification

30. A 37-year-old married man is diagnosed with HIV following unprotected sexual intercourse during his recent business trip to the Bahamas. The patient is reluctant to disclose the diagnosis to his wife out of fear that she might leave him. Which of the following is the most appropriate initial action by the physician?
- A) The physician should call the patient's wife and explain the situation
 - B) The physician should encourage the patient to discuss his diagnosis with his wife
 - C) The physician should report the case to the department of health so they can monitor the outbreak
 - D) The physician should report the case to the department of health so they can notify the wife of the possible exposure
 - E) The physician should not report the case due to the patient's right to privacy

The correct answer is B.

The department of health will be notified about the patient's diagnosis. However in terms of the physician's most appropriate initial action, it is recommended that he or she first encourage the patient to share the diagnosis with his wife. A plan to follow up with the patient should be negotiated. If the patient refuses to voluntarily notify his wife, involuntary notification may occur via the department of health, soliciting the testing of the wife.

The sequence for notification of partners of notifiable diseases is as follows:

Establish the diagnosis in the patient – e.g., chlamydia, gonorrhea, syphilis, HIV.

Counsel patient – explain nature of infection, how it is transmitted, the range of symptoms, possible complications, treatment, and follow-up.

Motivate patient to notify partners – explain the importance of informing contacts and preventing spread.

Negotiate plan for sexual contact – ask about current and past partners and agree who will inform each one. Make a record of each contact with names, contact details, and plans for how to contact.

Monitor progress – follow up with the patient about whether contacts have been notified.

Minimum necessary disclosure

31. A 40-year-old man presents to the emergency department with severe chest pain, shortness of breath, and cough productive of sputum. Further evaluation reveals pneumonia and the patient requires ICU hospitalization. The patient reveals that a few days earlier he got drunk at a party and lost consciousness. He also remembers vomiting several times that day. Following antibiotic therapy, the patient's condition improves and he is moved to the general ward. The patient's employer contacts the physician to ask about the patient's current state of health and his ability to return to work. The patient has authorized the release of all information. Which of the following information should the physician provide?
- A) A date when the patient is likely to be fit to return to work
 - B) A date when the patient is likely to be fit to return to work and a recommendation to keep him away from alcohol
 - C) Diagnosis and a date when the patient is likely to be fit to return to work
 - D) Diagnosis, a date when the patient is likely to be fit to return to work, and a recommendation to keep him away from alcohol
 - E) No information can be released

The correct answer is A.

Health Insurance Portability and Accountability Act (HIPAA) requires “minimum necessary” disclosure, meaning that the physician should disclose the minimal amount of information that is necessary (for the employer in this case).

Patient confidentiality should be maintained even if he or she has signed the release of information. No information can be shared without the consent of the patient.

There is no need for the employer to know the exact diagnosis. While the patient should be advised to abstain from unhealthy amounts of alcohol, notifying the employer of this recommendation would be inappropriate. The physician should only notify the employer of the patient’s fitness to return to work. No additional information should be shared.

Issues related to organ transplantation

32. A 45-year-old man with no significant past medical history has suffered diffuse axonal injury in a motor vehicle accident. Upon arrival at the emergency department he is pronounced brain-dead. He had previously expressed his desire to be an organ donor and his family has no objections. Which of the following would be an appropriate next best step in the organ donation process?
- A) Ensuring that brain-death is pronounced by the members of the transplant team
 - B) Ensuring that the members of the primary medical team that provided end-of-life care are not directly involved in retrieving or transplanting organs
 - C) Ensuring that transplant procedures are undertaken by the patient's primary medical team
 - D) Postponing the procedures if there are no potential recipients within the state
 - E) Transferring the patient to the facility where the potential recipients are hospitalized

The correct answer is B.

The American Medical Association (AMA) provides ethics guidelines for organ transplantation. The medical team involved in providing end-of-life care and pronouncing the time of death should not be the same as the team involved in organ transplantation due to potential conflicts of interest.

It is important that the staff involved in the organ transplantation process are educated and that it is performed at an adequately equipped facility. The organs are a national, rather than a state or local resource, and can be transferred as needed.

33. A 45-year-old man presents to the clinic for a physical examination prior to consideration for kidney donation. He would like to donate one of his kidneys to a former colleague in need of a transplant in exchange for a large sum of money. Which of the following is the most appropriate response by the physician?
- A) A compensation equivalent to the cost of the medical procedure only is acceptable
 - B) A compensation equivalent to the cost of the medical procedure and travel reimbursement only is acceptable
 - C) A compensation agreed upon personally by the recipient and donor is acceptable
 - D) The patient may not sell his kidney but may attempt to donate it

The correct answer is D.

Federal law prohibits the purchase or sale of organs from deceased and live donors.

Such payments have been prohibited since Congress passed the National Organ Transplant Act of 1984.

Gift law requires that any donor gifting an organ do so without monetary receipt.

Although the cost of the medical procedure and travel reimbursement may sometimes be covered through insurance or institutionally, direct monetary exchanges between recipient and donor are not considered ethical.

34. A 37-year-old man asks the physician if he and his wife could make some money selling tissues or organs. Which of the following would not be unethical to sell?
- A) Fertilized eggs
 - B) Heart
 - C) Kidney
 - D) Sperm
 - E) There are no tissues that can be legally sold

The correct answer is D.

There are no laws prohibiting the sale of sperm or unfertilized eggs. This is classified as compensation for gamete donation.

However selling fertilized eggs (embryos) and organs is illegal.

Reporting abuse

35. A physician is being investigated for failing to report a case of potential abuse. The physician's defense claims that the patient did not want him to report the case. In which of the following circumstances would the physician likely win the case?
- A) 16-year-old boy who got an A in his music class after having sexual intercourse with his teacher
 - B) 19-year-old woman sexually assaulted by her coworker
 - C) 35-year-old with Down syndrome and IQ of 40 physically abused by the carer
 - D) 70-year-old denied food by her son-in-law
 - E) The physician will lose the case because all abuse needs to be reported

The correct answer is B.

Mandatory reporting of suspected or confirmed abuse applies to situations involving minors (under 18), elderly (60+), and/or persons with disabilities.

Domestic abuse among adults does not require mandatory reporting. Patients should be supplied with a full spectrum of information (e.g., pamphlets, hotlines) and counseling. Treatment of victims of domestic abuse is aimed at preventing repetitions of violence and may require a combined effort between social services and healthcare professionals.

Bottom line: It is mandatory to report abuse of minors, elderly, and persons with disabilities. Domestic abuse among adults should be approached from an angle of providing information and resources to the abused, rather than reporting the abuse.

Child abuse

36. A 2-year-old girl is brought to the emergency department by her father due to pain and refusal to move her left leg. The father says that the girl was running down the stairs when she fell. X-ray confirms a spiral fracture of the left femur. During the examination, the girl warmly engages her father. Which of the following is the most appropriate next best step by the physician?
- A) Liaison with pediatrics and/or psychiatry for multiple opinions
 - B) Hospitalize the child and call child protective services
 - C) Hospitalize the child and organize a family meeting to discuss the injuries in more detail
 - D) Hospitalize the child, treat the injuries, and refer the father for parenting classes
 - E) Treat the injury and discharge the patient with close outpatient follow-up

The correct answer is B.

Abuse of minors, elderly, and persons with disabilities requires mandatory reporting.

Signs of child abuse include (but are not limited to):

- Spiral fractures (rotational force applied to bone) are pathognomonic for child abuse.
- Fractures at multiple stages of healing or in infants not yet able to mobilize.
- Circular burns (cigarettes).
- Scald burns sparing flexor areas (being dipped into hot water).
- Bruises on the shoulders, back, face, or dorsums of the hands (knees/shins are normal).
- Posterior rib fractures (being grabbed and squeezed).
- Retinal detachments or hemorrhages (shaken baby syndrome).
- Subdural hematomas (shaken baby syndrome).
- Avoidance of eye contact.

37. A 1-year-old boy is brought to the emergency department by his mother due to scald burns of his lower extremities. She says he fell into a bathtub full of hot water. The physician suspects the child has been intentionally submersed. In addition, the physician notices what appears to be a circular burn on the child's arm. Which of the following is the most appropriate response by the physician?
- A) "Can you please tell me in more detail exactly what happened?"
 - B) "Did you dip your child into hot water?"
 - C) "Has your child been babysat recently by someone other than yourself?"
 - D) "I know sometimes certain aspects of parenting can be difficult. Is there anything you'd like to talk about?"
 - E) "We will need to admit your child to hospital for additional care."

The correct answer is E.

The child needs to be admitted to hospital and child protect services contacted.

The parent should not be directly confronted about the potential abuse since he or she may flee with the child, thereby putting the child at risk of further harm.

A caregiver will almost never admit the truth of his or her actions. If the physician suspects abuse, asking for further detail is not the most appropriate step. The physician must admit the child and contact child protect services.

If the caregiver leaves with the child despite the physician's attempt to admit him or her, the next step is still to contact child protective services.

Domestic violence

38. A 35-year old woman presents to the emergency department with chest tenderness and pain on inspiration. She reports that earlier this morning she was walking around the house and “walked into a fridge.” The patient was recently hospitalized for vulvar bruising and thigh lacerations that she attributed to falling from a chair. On examination, a large bruise is noted on the right side of her chest. Chest x-ray confirms fractures of the 3rd and 4th ribs. The patient mentions several times that she has a partner who loves her very much. Which of the following is the most appropriate response by the physician?
- A) “Can you tell me more about your relationship with your husband?”
 - B) “Has your husband been hitting you?”
 - C) “I am afraid this looks like domestic violence to me and I am legally required to report it.”
 - D) “I am glad to hear your husband loves you but these bruises suggest he is being rough.”
 - E) “I know it is difficult to talk about abuse but I assure you everything you say is confidential.”

The correct answer is A.

Unlike with elder/child abuse, cases of suspected domestic abuse between adults should not be reported without their consent. In this case, it is likely that the patient is being abused by an intimate partner and it needs to be explored further.

An open-ended question is the best way to initiate a discussion. Once the patient becomes more forthright about the situation, it may be appropriate to ask more direct questions.

Immediately confronting the patient's description of the events would unlikely make her reveal the abuse. If the patient does reveal the abuse, the next step would be to supply her with a spectrum of information (e.g., pamphlets, hotlines) and ensure her safety by identifying a place she can go to in the event of an emergency.

Abortion issues

39. A 35-year-old woman presents to the physician for bloating, nausea, and vomiting. A urine pregnancy test is positive. The patient would like to terminate the pregnancy since she already has three children and, as a single mother, feels she could not properly care for the fourth. Her physician feels strongly that abortion is immoral and his faith strictly forbids it. Which of the following is the appropriate response to the patient's request?
- A) "Abortion is not something we should consider in this case."
 - B) "Before we undertake any abortive intervention, there is a required one-week waiting period."
 - C) "I am sorry but I do not perform abortions. I will refer you to another provider who does."
 - D) "While I personally disagree with abortions, it is my duty to perform one if you so wish."
 - E) "Yes, I can carry out a termination of pregnancy for you if you wish."

The correct answer is C.

Many physicians are opposed to abortions for a variety of reasons and do not wish to perform them. They are also not required to. Since this patient has the right to terminate her pregnancy, the physician should refer the patient to a physician who will carry out the management.

Following the 1973 United States Supreme Court landmark decision, *Roe v. Wade*, and subsequent companion decisions, abortion is legal in all 50 US states. It may however be restricted by the states to varying degrees. Some states have passed laws to restrict late-term abortions or have laws requiring parental notification for minors.

Participation in executions or torture

40. At the end of his shift at the state prison, a physician is asked to assist the prison personnel in an execution of an inmate who has been sentenced to death for a mass murder. The inmate is a known intravenous drug abuser and the prison staff anticipate difficulties obtaining adequate IV access to administer the lethal substances. They believe the assistance of a physician would reduce the risk of the inmate suffering. They also need the physician to prescribe a sedative a few hours before the execution to relieve the inmate of anxiety and to certify death once the inmate's heart stops. Which of the following would be the most appropriate actions by the physician?
- A) Prescribe diazepam; certify death; refuse to start the IV
 - B) Prescribe diazepam; certify death; start the IV
 - C) Prescribe diazepam; refuse to certify death until it is pronounced by another person; start the IV
 - D) Prescribe diazepam; refuse to certify death until it is pronounced by another person; refuse to start the IV
 - E) Refuse everything

The correct answer is D.

Physician participation in executions or torture in any form is not allowed. The physician may not advise the execution team on how to perform the execution, mix the drugs, or start the IV. The physician may also not certify the death immediately after it has occurred, but only after the inmate is pronounced dead by another member of the team. Prescribing diazepam to relieve anxiety prior to the execution is acceptable as this is aimed solely at relieving suffering and is in no way participating in the actual execution. It is also unethical for physicians to participate in torture or to use their knowledge or skills in any way that would facilitate it, regardless of circumstances and who orders it.

Obtaining information

41. A 76-year-old woman with a past medical history significant for hypertension, type II diabetes mellitus, congestive heart failure, and osteoarthritis presents to the clinic for the third time in two months because of blood pressure and glucose irregularities. Medicatoinis include insulin, metformin, metoprolol, lisinopril, spironolactone, hydralazine, isosorbide dinitrate, and acetaminophen. The physician suspects the patient has been having difficulties managing an increasing number of medications. Which of the following is the most correct way to initiate this conversation with the patient?
- A) "How do you feel about how we are medically managing your health?"
 - B) "How often do you skip your medication doses?"
 - C) "I am afraid you are not being compliant."
 - D) "People often struggle with so many medications. How has this been for you?"
 - E) "Would you like me to call your children to help you with medications?"

The correct answer is D.

It is helpful to initiate a discussion by acknowledging the difficulty the patient may be facing and asking an open-ended question, allowing the patient to express her thoughts in her own words. This can then be followed by more detailed and direct questions. If the patient says that she has been struggling to take her medications regularly, then possible solutions can be explored, such as “negotiating a contract,” which means writing things down (on NBME).

It is important not to come across judgmentally, which could make the patient defensive and less likely to disclose any issues she may be having. The suggestion of involving the patient’s children is not immediately appropriate at this time as there may be other causes of poor glycemic control and blood pressure regulation.

Delivering bad news

42. A 66-year-old man comes to the clinic for a follow-up appointment after having undergone an abdominal CT last month in the setting of painless jaundice and weight loss. The patient is a retired bus driver who has never had significant health problems. The CT shows a mass in the head of the pancreas. Upon entering the room, patient immediately asks, "What did the scan show?" Which of the following is the appropriate way to start the conversation?
- A) Ask the patient about his understanding of why he had to undergo the scan
 - B) Ask the patient if he had any advanced directives
 - C) Ask the patient to call a family member
 - D) Briefly describe the findings on the scan
 - E) Take additional history

The correct answer is A.

When delivering bad news to a patient, one of the most important initial strategies is to ensure the setting for communication is appropriate and the patient feels comfortable. When initiating a conversation, it is advisable to establish the patient's baseline understanding of need for testing so that the rest of the conversation can be adapted to accordingly.

While it may be beneficial to the patient to have a family member there with him during the delivery of bad news, the patient has asked point-blank what his CT showed. It is therefore the responsibility of the physician to inform the patient of the news, sensitively and empathetically. It is not mandatory to break bad news with family members present.

Briefly describing the findings of the CT without first establishing the patient's understanding of his need for the scan does not establish a communicative and empathetic framework for delivering bad news.

Taking additional history at this moment is dilatory and circumvents the delivery of bad news.

43. A 32-year-old female athlete presents to the clinic because of a recently palpated mass in her right breast. Further examination reveals a 3x3-cm, irregular, firm, immobile mass in her right breast, and two enlarged, painless lymph nodes in the ipsilateral axilla. Following investigation, the patient is diagnosed with regionally advanced breast cancer. The patient refuses to believe the diagnosis because she is young, symptom-free, and is an award-winning athlete. Which of the following is the best way to respond to this patient's denial of her diagnosis?
- A) "I'd like to discuss with you what we will do from here."
 - B) "I know you are very athletic. You're right that this could be benign after all."
 - C) "It must be difficult to hear such news when you are feeling so well."
 - D) "Look, everything will be okay."
 - E) "What do you want to do from here?"

The correct answer is C.

The physician must acknowledge the patient's feelings of denial in order to establish rapport and engage in meaningful discussion. It is common for patients who feel healthy to struggle to grasp the gravity of their diagnosis. Delivering bad news requires establishing the patient's baseline level of understanding of the situation through empathetic statements, which are then followed by a more detailed explanation by the physician. At each step, the patient's feelings must be acknowledged and respected. She should be given sufficient time to process the information. It would be inappropriate to interrogate the patient or to criticize her, just like it would be inappropriate to say that everything is going to be fine because the outcome is impossible to know at this point.

The five Elisabeth Kübler-Ross stages of grief are:

- Denial
- Bargaining
- Anger
- Lamentation
- Acceptance

44. A 15-year-old boy with no significant past medical history presents to the clinic because of discomfort in his right testicle. Following investigations, a diagnosis of invasive seminoma is made. The physician recommends orchiectomy plus radiotherapy. The boy says he wants to undergo full treatment. The mother consents to both the radiotherapy and orchiectomy. The father consents to radiotherapy only. Which of the following is the most appropriate next best step?
- A) Arrange for a court to oversee the case
 - B) Arrange for radiotherapy and orchiectomy
 - C) Arrange for radiotherapy only
 - D) Attempt to reach further consensus between the parents
 - E) Withhold management because radiotherapy only is futile

The correct answer is D.

Children cannot independently give consent to their own cancer treatment. In the event that parents disagree regarding the nature of how management should proceed, an attempt to establish further consensus should be sought.

If the father ultimately refuses the orchiectomy component for his son, this procedure cannot be performed. The question as to whether to proceed with radiotherapy only in this case would be up to the discretion of the physician.

When parents refuse cancer treatment, there are usually four main reasons:

- 1) They prefer alternative and/or complimentary medicine.
- 2) Conflicts in relation to faith.
- 3) Concern about adverse effects.
- 4) Lack of insight regarding the patient's treatment needs.

Clinicians can usually reach resolutions with parents by listening to their concerns and engaging in informative discussion with disclosure of all risks and benefits. Clinicians may involve secondary consults with psychologists, ethicists, chaplains, or the palliative care team. A multi-disciplinary approach may help all parties reach a consensus. However, ultimately, the parents have the right to refuse any or all cancer treatment for their child who is a minor.

The after-hours patient

45. A 35-year-old woman with a history of making unnecessary emergency appointments presents to an outpatient clinic after-hours. She insists on being seen immediately for a rash that she has had for the past five days. She apologizes for showing up unannounced and claims she is a very busy lawyer. Which of the following is the most appropriate response by the physician?
- A) "I will check it out but in the future please try to schedule appointments during business hours."
 - B) "If you believe it is an urgent matter, I will see you now."
 - C) "Please schedule an appointment during business hours."
 - D) "There is an on-call physician in a nearby office who can see you right now."
 - E) "This kind of behavior is inappropriate."

The correct answer is C.

It is important for physicians to set boundaries with patients. This patient is inappropriately demanding to be seen after-hours for a non-urgent matter. The physician should be firm, but polite, and explain to the patient that she should respect the business hours of the practice and make an appointment. Examining the patient after-hours would only encourage similar behaviors in the future.

Referring the patient to an on-call physician would be inappropriate since this can be resolved during office hours.

It would be inappropriate to refer her to the emergency department for something that is unlikely to be a medical emergency.

Telling a patient that her behavior is inappropriate would damage the physician-patient relationship.

If the physician believes the situation is a genuine medical emergency, he or she can choose to manage the patient or see that she receives appropriate care at an emergency department. Physicians are not required to see patients after-hours.

Physician disagreement

46. A 75-year-old woman is being treated for amyotrophic lateral sclerosis. Her functional status has deteriorated significantly and her prognosis is <6 months of life remaining. Her neurologist recommended she be transferred to hospice care. However, during her last hospitalization for pneumonia, another neurologist recommended her for a clinical trial and believes that a new stem-cell based treatment could prolong her survival. The patient says she just wants what is best for her. Which of the following is the appropriate next step by her primary care physician?
- A) Ask the patient how she feels about hospice care and the clinical trial
 - B) Contact the two neurologists and try coordinating care
 - C) Refer the patient for hospice care
 - D) Refer the patient for the clinical trial
 - E) Request the opinion of a third neurologist

The correct answer is B.

One of the roles of the primary care physician is to coordinate care. In this case, the two neurologists disagree on the course of action for this particular patient. The primary care physician should not judge the competence of two specialists by choosing which of their opinions to trust but instead try reaching out to them and seeing whether an agreement can be reached. It is possible that the reason the physicians disagree is that one is not aware of the clinical trial.

The final decision for any treatment is made by the patient as per her autonomy, but the primary care physician should first try to reach an agreement between the neurologists as the next best step.

Bottom line: When disagreements arise between healthcare providers, one of the roles of the primary care physician is to coordinate care for the patient by enhancing communication between parties.

Obtaining consent

47. A 55-year-old woman is about to undergo elective cholecystectomy. While obtaining consent, she is informed of the rationale behind the procedure, the basics of the operating technique, alternative treatments, and complications. She is given a list of all medications that will be used during the procedure and informed that a resident surgeon will also be present in theater. Which of the following could have been omitted while obtaining consent?
- A) Alternative treatments
 - B) Complications
 - C) Medications used during the procedure
 - D) Presence of a resident surgeon
 - E) Surgical technique

The correct answer is C.

During the process of obtaining consent, the patient must be told why the procedure is to be done, how it is done, what possible complications there are, and whether any alternative treatment options exist. She also needs to know who the physician performing the procedure is and if other staff may play a role in the surgery. It is not necessary to provide a list of all medications used during the procedure, although her relevant history of allergies and past/current medications are paramount to know.

48. A 35-year-old man has been hospitalized for acute pancreatitis. Over the next few days he develops a large pseudocyst. Internal drainage via ERCP is planned. The resident on the team has no experience with this procedure. The nurse has assisted the attending physician a number of times during this type of procedure. It would be appropriate for which of the following team members to obtain consent for the procedure?
- A) Attending physician and the nurse
 - B) Attending physician and the resident
 - C) Attending physician only
 - D) Attending physician, the resident, and the nurse
 - E) This procedure does not require additional consent once the patient has consented to pancreatitis management

The correct answer is C.

Consent should be obtained by the team member most familiar with the procedure who can answer any questions the patient may have. While it may be acceptable for medical students, residents, and support staff to accompany the attending physician while obtaining consent, in order for the patient to be fully informed, the attending him or herself must obtain the consent.

It would be inappropriate to ask the nurse to obtain consent. Likewise, the resident does not have sufficient knowledge of the procedure to be able to fully inform the patient. Consent is necessary for surgical interventions. Just because the patient consented to hospitalization and pancreatitis management does not automatically mean he consents to pseudocyst drainage via ERCP.

Language-related issues and interpreters

49. A 75-year old woman presents to the clinic because of unstable blood pressure. She is accompanied by her 45-year old daughter. The patient is an immigrant from Guatemala and speaks limited English. Her daughter is comfortable being the interpreter for her mother. The physician's knowledge of Spanish is limited. Which of the following is the best approach to this situation?
- A) Allow the daughter to translate
 - B) Attempt to speak the patient's native language
 - C) Provide all information in written Spanish
 - D) Speak slowly and use images to illustrate ideas
 - E) Use an official interpreter

The correct answer is E.

When a situation is non-urgent and an official interpreter can be reached, either in-person or over the phone, this is always preferable over having family members carry out translations. This is because family members may inadvertently alter the meaning of important communication, or a patient may not wish to have the family member involved in certain parts of the communication process.

Speaking slowly and using images may be advisable alongside providing written information, however it is important to communicate to the patient in a language she can understand to reduce the likelihood of medical error.

It would be acceptable for a family member to translate in an emergency when an official translator cannot immediately be reached.

A physician should not use a language he or she is uncomfortable with since miscommunication can have serious consequences.

Bottom line: Many patients will be perfectly willing to have family members carry out translations, but in non-urgent situations the most appropriate step is to seek an official interpreter.

Ethical principles of medicine

50. A 68-year-old woman with metastatic lymphoma presents to the emergency department with fever, dysuria, and back pain. She is diagnosed with pyelonephritis and antibiotics are prescribed. The patient, however, wants to fight through without antibiotics because it's going to be a full moon tonight and that will certainly give her strength. Her husband arrives at the hospital and wants her to receive antibiotics now. Which of the following is the most appropriate next step in managing this patient?
- A) Administer intravenous ampicillin and gentamicin
 - B) Assess the patient's decision-making capacity
 - C) Consult the hospital ethics committee
 - D) Discharge the patient as per her right to autonomy
 - E) Seek a court order giving the husband the legal power to make medical decisions

The correct answer is B.

This is a tricky question because the patient's autonomy must absolutely be respected, however her decision-making capacity should first be determined. This is not unique to this situation. Any time a patient is treated, decision-making capacity should be assessed and informed consent obtained.

Since the patient does not want antibiotics, she will be discharged without them. But before this is done, her decision-making capacity will still be assessed. It can then be appropriately documented alongside that the patient chose to be discharged against medical advice.

If a patient refuses medical care, implement patient education, understanding, and informed consent. Explore reasons behind the refusal. Then document the patient's refusal.

The husband's desire for the patient to receive antibiotics is irrelevant. Only if the patient becomes incapacitated and he is her durable power of attorney would he have the ability to override her choice to not receive antibiotics.

51. A 45-year-old man presents to the clinic for a routine follow-up examination. Blood pressure has been >150/100 mmHg on his previous three appointments. The physician recommends lisinopril to manage the patient's hypertension. After a lengthy discussion, the patient declines medications. This case best describes which of the following ethical principles?
- A) Autonomy
 - B) Beneficence
 - C) Justice
 - D) Morality
 - E) Non-maleficence

The correct answer is A.

Autonomy refers to a patient's ability to accept or refuse medical treatment. It is very rare for interventions to be made against patients' wishes, even if decision-making capacity is dubious. In this case, the patient has elected not to take the recommended medication, thereby exercising his right to autonomy.

Beneficence refers to the physician's duty to do what is best for the patient. "Best" may be derived from a combination of the physician's judgment and the patient's wishes. In this case, the physician's desire to prescribe an ACE inhibitor is acting out of beneficence – i.e., he is recommending what he believes is best for the patient.

Non-maleficence refers to not doing harm to a patient. An example would be reducing a patient's unnecessary polypharmacy. Another example would be to refuse to provide certain treatments or interventions (e.g., prescribing high-dose iron tablets instead of an ACE inhibitor because the patient has heard iron is beneficial).

Justice refers to fairness in resource distribution, where patients in similar situations should have the same access to care.

52. A 10-year-old boy is diagnosed with terminal acute lymphoblastic leukemia. His parents do not want him to know his prognosis. The boy is in the 9th grade and has an IQ of 160. He asks to speak alone with the physician briefly. His parents acquiesce. Once in private, the boy says, "Look, I know my parents just want what's best for me and they're looking out for me. These types of communications are important and sensitive, I understand. Can you just be honest with me: am I dying? I am an adult and bear my own autonomy." Which of the following is the most appropriate response by the physician?
- A) "An inverse relationship is observed between lymphocyte count and prognosis in your case."
 - B) "Have you spoken to your parents about this?"
 - C) "I cannot discuss your prognosis with you without your parents present."
 - D) "You will be okay. These types of things are stressful for your parents."
 - E) "Your prognosis is not good, no. You must understand your parents are stressed."

The correct answer is B.

This is a favorite USMLE question. In these types of scenarios where the minor privately confides in the physician, yet the parents do not want prognosis conveyed, the physician's best response is to ask the child how communication has been with his or her parents. The discussion should be as open-ended as possible.

Choice C does not sound unreasonable, but it is not as ideal as first assessing the nature and scope of the child's communications with his parents. Choice C may place the parents in an uncomfortable position, where the child may believe information is being hidden from him.

Choice D would be lying to the patient.

Choice E is wrong because parents have the right to refuse their children being aware of prognosis.

53. A 75-year-old man presents to hospital with severe epigastric pain. An abdominal x-ray reveals air under the diaphragm. Surgery is recommended. The patient, however, despite being fully informed, refuses the surgery and requests conservative management only. Soon afterward, his condition deteriorates and he loses consciousness. His wife and both adult children ask for the surgery to be performed. Which of the following is the correct action by the physician?
- A) Ask the family members why they would like to overrule the wishes of the patient
 - B) Continue with conservative management
 - C) Do the surgery
 - D) Seek a court order to do the surgery
 - E) Refer the case to the hospital ethics committee

The correct answer is B.

Patients have the legal right to refuse treatment even if it is life-saving. The patient was fully informed and competent at the time when he made it clear he did not wish to have surgery. His wishes need to be respected as per the ethical principle of autonomy.

When the patient is incapacitated and his wishes are not known, his next of kin can act as surrogate decision makers (spouse before adult children). In this case, the patient's wishes are known.

Referring the case to the hospital ethics committee or the courts would be inappropriate. The hospital ethics committee can get involved when the patient's wishes are not known and the family members disagree on what course of action the patient would choose.

As discussed earlier, if the patient has a durable power of attorney, then if the patient becomes incapacitated, the wishes of the durable power of attorney must be followed, even if the patient stated views to the contrary prior to incapacitation.

54. A 40-year-old male presents to the clinic with weakness, night sweats, and weight loss. He has history of sexual intercourse with both genders. Initial workup reveals leukopenia. Screening for HIV is discussed with the patient and an ELISA is recommended. The patient declines the ELISA but says he is open to doing a PCR test. Which of the following is the most appropriate next step by the physician?
- A) Arrange for a PCR for the patient
 - B) Do not perform the PCR because it is not recommended for initial screening
 - C) Do not perform the ELISA
 - D) Explain the purpose of the ELISA test
 - E) Transfer care to another physician

The correct answer is D.

The patient has the right to refuse any type of medical care, as per autonomy. The USMLE will sometimes ask questions that seemingly have two “correct” answers. Choice C is not wrong in the sense that the physician must not perform an ELISA on a patient who refuses. However prior to fulfilling the patient’s wishes, out of the principle of beneficence, the physician must first explain the purpose of the ELISA test and document the patient’s rationale for declining the test.

The possibility of an HIV PCR test may be explored for the patient, but not before first explaining the purpose of the ELISA screening test.

55. A 27-year-old woman at 39 weeks gestation is in a car accident and bleeding profusely. Fetal monitoring shows late decelerations. Emergent Caesarean delivery is recommended to save the patient and fetus. The patient refuses. The husband arrives on the scene. He is screaming that he wants everything possible to be done to save his wife and unborn son. He says his wife is in a state of shock and not in a position to refuse treatment. He is a wealthy lawyer and says he will bring down the physician and the hospital if they don't act now. Soon afterward the patient loses consciousness. Which of the following is the appropriate action by the physician?
- A) Do not perform the surgery
 - B) Obtain an emergency obstetric consult
 - C) Perform the surgery while maintaining care and supervision of the patient
 - E) Send the patient to emergency theater and immediately transfer care to obstetrics

The correct answer is A.

A fetus is legally considered part of a woman's body. Even if the fetus is term and could be delivered safely, this patient has refused surgical intervention and her wishes must be respected. She should be resuscitated to the best efforts of the physician team, but surgery cannot be performed since she rejected it. Autonomy must be respected.

This is no different from, e.g., a woman with placenta previa who refuses a Caesarean delivery, despite the risk of death to both her and the fetus being high.

If the mother becomes deceased and the fetus is still viable, surgery may be performed to save the fetus. However if the mother is alive and has refused surgery, the fetus is considered part of her body and, once again, her autonomy for no surgical intervention must be respected.

The husband's desire to save his wife and unborn child does not overrule the autonomy of the mother. She has the right to refuse surgery.

56. A recently married 25-year-old woman with a family history of Huntington disease was tested for the disorder during her previous visit. She was extensively counseled prior to the testing. She and her husband wish to have children. Today she presents to the clinic in order to discuss the results. However, she says that she has changed her mind and does not want to know the result. The physician knows that her result is positive. Which of the following is the appropriate response by the physician?
- A) "From a population health standpoint, we must discuss your testing."
 - B) "I do not need to tell you the results now. Can you explain why you don't want to know?"
 - C) "I understand."
 - D) "Is there someone else you want me to share the results with?"
 - E) "Let's arrange a follow-up."

The correct answer is C.

The physician should communicate that he or she understands, and then let the patient know that she can return to the clinic at any time to receive the results.

Patients have the right to refuse information just like they have the right to refuse testing or treatment. They do not need to provide justification for their desire to not hear information. Their wishes must be accepted. Learning about a potential gene mutation one might have does not necessitate mandatory disclosure from any type of public health standpoint.

In the event the patient has a communicable disease of public health significance (i.e., HIV, hepatitis C, etc.), the patient must be informed of his or her diagnosis in order to prevent community spread.

Patients can also request information be shared with someone else as opposed to directly receiving the news themselves. This possibility can be discussed with the patient, but not before communicating an understanding for the patient's wishes.

A follow-up appointment can be arranged for the patient, but first her desire to not know the test result must be respected by the physician.

57. A 56-year-old man with diabetes mellitus is participating in a clinical trial exploring the efficacy of a novel glycemic modulating agent. However he soon becomes bothered by the frequent monitoring of the trial and wants to cease his involvement. Which of the following is the most appropriate response by the physician?
- A) "I'm sorry you are bothered by the frequent testing. Would you like to reconsider?"
 - B) "No problem. However you will be ineligible for re-entry into the trial if you leave now."
 - C) "No problem. Please fill out this exit survey."
 - D) "You can quit."
 - E) "You can quit. To help other patients, would you mind explaining what you didn't like?"

The correct answer is D.

Patients can leave clinical trials at any point without the need to provide any line of justification. The decision of the patient must be respected.

The USMLE will often have many similar-sounding answers that don't seem unreasonable, but the ethics learning point here is that a patient has the autonomy to leave a study/trial, period. A discussion about the trial may be engaged upon if the patient voluntarily and willingly discusses his or her thoughts. But if he or she communicates a desire to leave, albeit a terse justification or not, it must be respected.

58. An 88-year-old man with history of chronic kidney disease deteriorates and his physician recommends hospice care. However, the patient replies that he does not want to discontinue his dialysis. Which of the following is the most appropriate response by the physician?
- A) "Dialysis may be continued outside of hospice care."
 - B) "Hospice care is suitable in this case. Unfortunately this means stopping routine dialysis."
 - C) "We will not stop your dialysis in the hospice setting."
 - D) "What is your understanding of palliative versus hospice care?"
 - E) "Would you mind please explaining your understanding of dialysis?"

The correct answer is A.

Dialysis is considered life-extending treatment. It is also very costly and time-consuming to undergo. It is rare for hospice arrangements to include any type of dialysis as it is not considered a mere end-of-life comfort measure.

The eligibility criteria for hospice care are that the patient is terminally ill within six months and is not pursuing any type of life-prolonging management.

Palliative care may be explored as an option in which dialysis can be pursued. At this moment, asking the patient about his understanding of palliative versus hospice care, the latter which would require the cessation of his dialysis, is inappropriate, as the patient has already voiced that he wishes to remain on dialysis.

The patient does not need to explain his understanding of dialysis in order to continue receiving it.

Infectious disease and surgery

59. A 45-year-old woman with HIV presents to the clinic with suspected cholelithiasis. An abdominal ultrasound reveals gallstones and the patient is referred to a surgeon for cholecystectomy. The surgeon refuses to accept the patient due to her HIV status. Which of the following is the most appropriate next step?
- A) Explain to the patient that the surgeon does not feel comfortable treating her
 - B) Explore non-surgical options for the patient that are equally as efficacious as cholecystectomy
 - C) First measure the patient's viral load, then ask the surgeon to reconsider
 - D) Perform the surgery using laparoscopic technique
 - E) Transfer care to another physician

The correct answer is E.

An on-the-job physician is allowed to refuse to give non-emergent care to patients under most circumstances if he or she does not feel comfortable providing it. This is inclusive of reasons such as infectious disease status, even if this justification might run contrary to ethical or societal norms. Care can be transferred to another physician without explaining the reason to the patient.

It is illegal for a physician to refuse to provide care to a patient for reasons of ethnicity, race, gender, religion, or sexual orientation.

Physicians are required to attempt emergent/life-saving care if they are on the job. Physicians who are not on the job are not mandated to provide exigent care.

Verbal consent

60. A 76-year-old man is brought to emergency incapacitated by paramedics. He was reportedly spotted lying on the street by a neighbor who called 911. Further investigation reveals that the patient has likely suffered from ruptured diverticulitis and is now in septic shock. He requires antibiotics and surgery. Which of the following is the most appropriate way to obtain consent in this case?
- A) Proceed with interventions without consent
 - B) The daughter can consent to both the antibiotics and surgery over the phone
 - C) The daughter can consent to the antibiotics but not for the surgery over the phone
 - D) The daughter needs to be asked to come to the hospital to provide consent
 - E) The neighbor can consent to the antibiotics but not the surgery

The correct answer is A.

The patient is incapacitated in a scenario requiring life-saving intervention. Physicians may take the appropriate emergency measures needed in this case to save the patient.

Obtaining consent from family members over the phone has a limited role in emergent medical care. In this scenario, the patient requires life-saving treatment. Consent from family members is not required here.

Consent granted by family members over the phone may have a time and place, e.g., parents consenting to non-emergent care of their children.

61. A 45-year-old woman presents to the clinic with her husband due to severe upper abdominal pain. She is diagnosed with cholecystitis and surgery is recommended. The procedure is explained and the patient agrees to it. However, when the physician brings the form for her to sign, the patient says that she comes from a very religious household and that her husband needs to sign the form for her because women in her family traditionally never sign anything. The husband is willing to do it. Which of the following is the most appropriate action by the physician?
- A) Ask more about the patient's cultural and religious beliefs
 - B) Ask the husband to allow his wife to sign the form
 - C) Consult the hospital chaplain
 - D) Document the patient's refusal to sign and allow the husband to sign
 - E) Document the patient's verbal consent and proceed with planned management

The correct answer is E.

Written consent is not essential. Verbal consent, while more difficult to prove, is equally valid. A second witness to oral consent is highly advisable in case it is ever brought into question. This patient has consented to the procedure. She is legally competent to make healthcare decisions so there is no need for surrogate decision-making or contacting the hospital chaplain. Every attempt should be made to provide culturally sensitive care to patients and demanding a signature would be inappropriate.

Although asking more about the patient's cultural and religious beliefs may be appropriate in some settings (e.g., if the patient asks about unusual or alternative therapies), in this case, acquiring verbal consent is acceptable.

Accepting gifts from pharmaceutical companies

62. A physician is asked to deliver a presentation on a novel agent sponsored by a pharmaceutical company. The physician accepts the invitation. The company offers to prepare a poster for the presentation, provide monetary compensation to the physician, and cover the travel costs of all attendees. Which of the following is considered ethically acceptable?
- A) All of the above
 - B) Compensation for the lecturing physician and the travel expenses of the attendees
 - C) Compensation for the lecturing physician only
 - D) Travel expenses of the attendees only
 - E) Travel expenses of the attendees and the cost of the poster production

The correct answer is C.

It is permissible for physicians invited to deliver presentations to be compensated for their time and reimbursed for travel costs. Full disclosure of any financial ties between the company and lecturer is essential. Moreover, if the physician participated in any way in the research project, this also needs to be disclosed.

It is not permissible to receive travel reimbursements for merely attending, but not presenting at, conferences, as this may preclude the ability of healthcare professionals to make impartial assessments.

It is important to remember is that the content of such presentations needs to be decided solely by the physician delivering the lecture. The pharmaceutical company should not be influencing the content (i.e., by providing slides or lecture notes).

63. A physician is attending a large cardiology conference sponsored by the pharmaceutical industry. A number of practitioners from different parts of the world are presenting their research. Inexpensive lunch is provided to all attendees. Everyone is also given a stethoscope. At the end of the day, one randomly chosen attendee is offered a one-week holiday in Hawaii, and the physician is notified that he is the winner. Which of the following is the most appropriate response by the physician?
- A) "Hawaii? Why not Barbados?"
 - B) "I cannot accept the holiday."
 - C) "I am unsure and would like to discuss the nature of this prize with the medical board first."
 - D) "Thank you."
 - E) "What more do I need to know before accepting this gift?"

The correct answer is B.

The issue of accepting gifts from pharmaceutical companies is complicated. It is generally agreed upon that it is acceptable to take only the smaller gifts that will directly help with patient care. An example would be educational material on healthy diets for patients with hypertension or diabetes. It is also acceptable to have the meal.

Extravagant gifts such as a holiday in Hawaii (or Barbados) might function to unduly influence the physician and would clearly violate ethical standards. In cases of physicians delivering lectures at events sponsored by pharmaceutical companies, the content of the lecture needs to be fully controlled by the physician, not the company sponsoring the event.

Choice C does not seem like an unreasonable answer, but this question is tricky in that physicians should know concretely that such gifts are not acceptable. In other words, tempting, but no.

Accepting gifts from patients

64. A 75-year-old woman comes to the physician for a follow-up examination following a successful hip surgery. She is extremely grateful that she is able to walk again and can play with her grandchildren. To express her gratitude, she brings the physician a basket of oranges she picked up at the grocery store. Which of the following is the most appropriate response by the physician?
- A) "Thank you for the oranges."
 - B) "This was very nice of you, but I cannot accept gifts from patients."
 - C) "This was very nice of you, but I cannot accept perishable food items here in the clinic."
 - D) "This was very nice of you, but I am allergic to oranges."
 - E) "This was very nice of you. Let me first consult with hospital management about whether it is okay for me to accept this gift."

The correct answer is A.

Accepting gifts from patients is a difficult issue because rejecting a patient's gift may leave him or her feeling embarrassed or shameful, inadvertently causing harm to the doctor-patient relationship.

It is generally considered acceptable for the physician to accept low-cost gifts as per his or her best judgment (e.g., simple food items or thank-you cards). If gifts are of notable monetary value however (e.g., sports tickets or a wristwatch), they must be firmly and politely rejected.

Romance between physicians and patients

65. While attending a medical conference, a physician is approached by a colleague. Following a brief conversation, both physicians develop a romantic interest in each other. They share a beer together and have an intimate encounter. Soon afterward, the colleague reveals to the physician that while she was in medical school, she attended a resident-run clinic and he treated her for a brief episode of major depression. She assures him that she is fine now and would like to continue the interaction. Which of the following is the correct response?
- A) "Alright."
 - B) "It is always inappropriate to date former patients."
 - C) "It is always inappropriate to date other physicians."
 - D) "I'm sorry, but we need to end our interaction."
 - E) "Since it was a long time ago, it is a possibility. I will consult the ethics board first."

The correct answer is D.

It is always unacceptable to date any current patients and former psychiatric patients.

While it is generally advisable to avoid dating any former patients, if the patient had been treated for a non-psychiatric condition, the relationship would be permissible if the physician-patient relationship had been terminated at least 6 months to 2 years prior.

There are no rules against physicians dating other physicians insofar as no ethical or legal rules are broken (e.g., sharing patient information, or inappropriately referring patients between each other to increase number of visits).

Prescribing for family members

66. A physician receives a call from his sister, a 45-year-old female with no significant past medical history who complains of sleep deprivation. She works as an attorney and has numerous important cases coming up, the outcomes of which determine whether she becomes a partner at her law firm. She says that her regular physician has always been giving her zolpidem but she has been unable to get an appointment on time. To the best of the physician's knowledge, his sister is reliable and has never abused any substances. She is asking for a single zolpidem prescription until she can schedule an appointment with her primary care physician. Which of the following would be the most appropriate response?
- A) "Can you first show me one of the previous prescriptions you were given?"
 - B) "Can you tell me more about why you think you need zolpidem in particular?"
 - C) "I can't write this prescription for you."
 - D) "Treating family members is never appropriate."
 - E) "You should make me your primary care physician first."

The correct answer is C.

This question is “hard” because this situation seems otherwise benign/innocuous, but the correct answer is that it is unacceptable to write a prescription in this case. The USMLE wants you to understand when it is important to take a professional hard-line.

Treating family members and friends is generally discouraged unless it is an emergency and no other physicians are available. In this non-urgent case, it would be inappropriate to prescribe medication both because the physician’s sister is not his patient and because she may not be completely honest with a family member.

She should instead be encouraged to make an appointment with her current primary care physician and discuss the concern with him or her.

Asking for additional details is unnecessary as it would still be inappropriate to prescribe zolpidem regardless. Naming a family member as one’s primary care physician can be problematic and should be avoided.

Impaired colleague

67. While on his ICU rotation, a resident notices that one of the attendings has been sleepy at work. One day, the resident notices the attending taking a couple of benzodiazepine pills during the shift. Soon afterward, while attempting to insert a central line, the physician punctures an artery causing a major bleed, necessitating a blood transfusion for the patient. Apart from this isolated incident, this resident is not aware of any potential mishaps by the attending. The attending is well-respected amongst his colleagues and is an esteemed staff member at the hospital. Which of the following is the appropriate action by the resident?
- A) Ask the attending if he believes his sleepiness is a concern
 - B) Ask the attending if he has been taking medications to help with wakefulness at work
 - C) No immediate action is needed as this is an isolated episode
 - D) Notify the attending that you will be reporting the concern
 - E) Report the concern without notifying the attending

The correct answer is E.

When noticing that a physician may be impaired and unable to perform his or her duties, it is a responsibility of other members of staff to report it since their primary duty is to the patients.

In this case, it is apparent this attending's actions are putting patients' well-being at risk. While the physician's use of benzodiazepines cannot be over-speculated and needs to be appropriately explored, at this point, the most important intervention is to prevent the physician from harming another patient.

The exact practices for reporting impaired colleagues are not consistent across all healthcare centers and institutions. Although in many instances colleagues are encouraged to directly discuss concerns with each other, this attending should be reported, and it is not required that he be notified first. Anonymous reporting is acceptable and ensures patients are maximally safe.

The degree to which the attending is professionally venerated is irrelevant in terms of the need to be reported.

Inappropriate behavior by another physician

68. A 30-year-old woman presents to the clinic for the first time to establish care. She used to see another physician previously but has decided to change clinics. She has no current complaints. Her past medical history is insignificant. The physician proceeds to discuss health maintenance and asks about history of Pap smears. Upon mentioning them, the patient starts crying. She reveals that the reason she is changing clinics is because her previous physician had been asking her to take off her clothes frequently during visitations and had been touching her. The previous physician insisted such examinations were not out of the ordinary. Which of the following is the most appropriate next best step?
- A) Ask the patient for more details in order to first better ascertain the situation
 - B) Call the previous physician and ask about the patient's history without bringing up the patient's complaints
 - C) Call the previous physician and address the patient's complaints
 - D) Report the previous physician
 - E) Tell the patient that you are concerned and will report her previous physician

The correct answer is D.

The previous physician must be reported and the issue explored by the medical board. The patient does not need to be notified that you will be making the report. This may also put her in an uncomfortable position since she is attempting to confide in you.

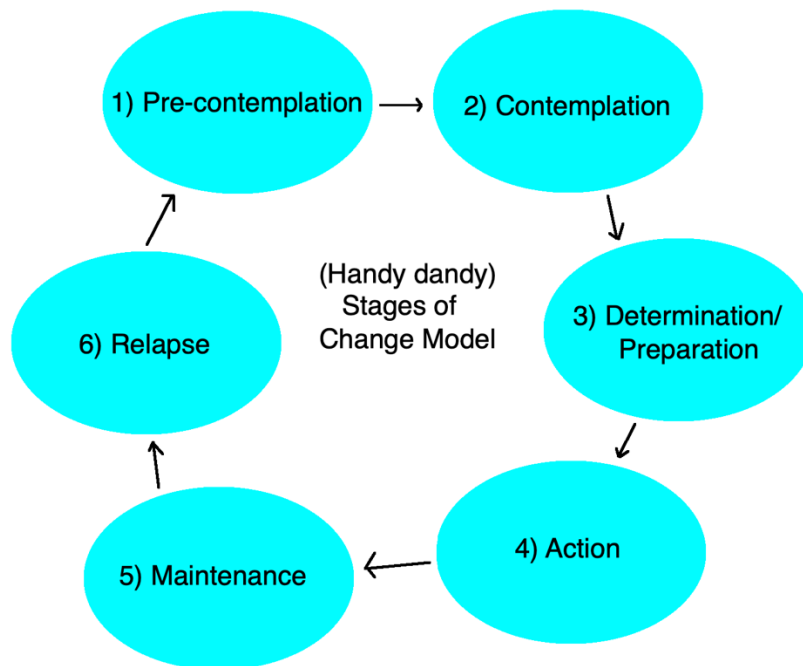
Although it might sound reasonable to ask further questions so that one can better ascertain whether or not to report the previous physician, this is not the most appropriate step because the reporting needs to be done regardless. In other words, the patient has objectively changed clinics and broken down crying saying she was concerned about her previous physician. This alone warrants reporting.

The previous physician need not be contacted regarding the patient's complaints.

Habit reversal

69. A 60-year-old man presents to the clinic with shortness of breath on exertion and a cough productive of sputum for the past three months. He has a 40-pack-year smoking history. The physician wishes to begin a discussion about smoking cessation. Which of the following would be an appropriate way to initiate the conversation?
- A) "Have you ever considered nicotine patches or electronic cigarettes?"
 - B) "Have you ever thought about smoking cessation?"
 - C) "What do you think about the role smoking plays in the development of lung cancer?"
 - D) "You don't have to quit smoking. However we should quickly discuss how things could improve if we cut back."
 - E) "Your health would improve significantly if we can get you to quit somehow. What do you think?"

The correct answer is B.



We need to ask an open-ended question to assess what stage of change this patient is in with regard to smoking cessation. This is the best initial step. The patient does not need to be lectured about harmful effects or imposed upon with methods to quit. The patient should be made aware that he or she is in full control, and the physician should do his or her best to not communicate from a place of condescension or judgment.

Pre-contemplation: The patient has not yet considered (i.e., contemplated) the notion of changing his or her behavior.

Contemplation: The patient has considered (i.e., contemplated) the idea of changing his or her behavior but has not yet taken any concrete or tangible steps to implement such change.

Determination/Preparation: The patient is mentally committing to change but has not yet reduced or ceased the behavior (e.g., reading pamphlets/websites about harmful effects and plans to start the nicotine patch).

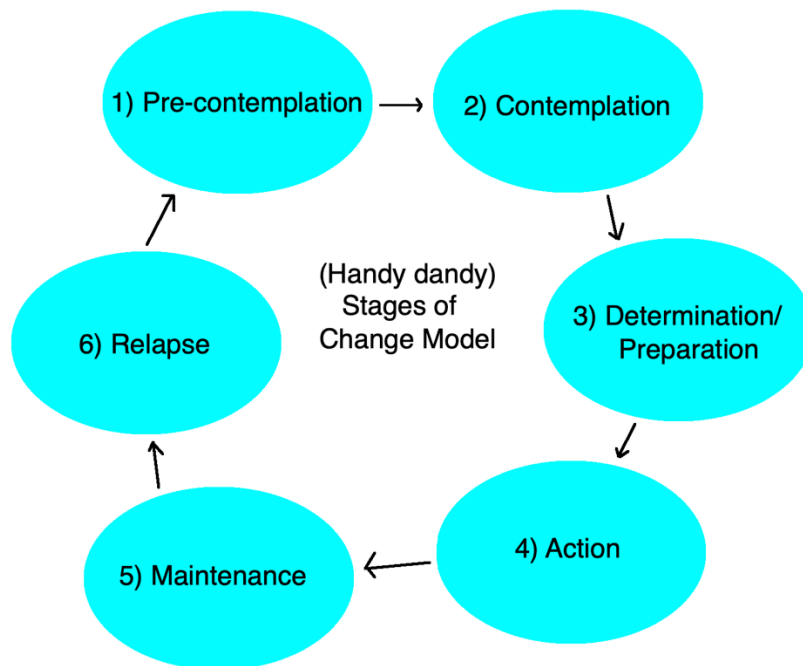
Action: The patient is now taking concrete steps to reduce or stop a behavior (e.g., has commenced nicotine gum/patch). There is minimal stability to the patient's behavior at this point and chance of relapse is high.

Maintenance: The patient has stably implemented behavior to reduce certain behaviors (e.g., has been off of cigarettes for three weeks and is currently on the patch).

Relapse: The patient has experienced a recurrence (i.e., relapse) of the behavior (e.g., was on the patch for a month but has resumed occasional smoking).

70. A 45-year-old man presents to the emergency department with hematemesis. The patient is a known alcoholic with confirmed esophageal varices. After the patient is stabilized and ready to be discharged, he acknowledges that his alcohol use has caused these problems but says that it is probably too late now to quit anyway. Which stage of change best describes this patient?
- A) Action
 - B) Contemplation
 - C) Maintenance
 - D) Pre-contemplation
 - E) Preparation

The correct answer is B.



This patient with advanced cirrhosis and esophageal varices has contemplated the notion of changing his behavior, albeit adhering to the sentiment that quitting alcohol this late is futile.

Pre-contemplation: The patient has not yet considered (i.e., contemplated) the notion of changing his or her behavior.

Contemplation: The patient has considered (i.e., contemplated) the idea of changing his or her behavior but has not yet taken any concrete or tangible steps to implement such change.

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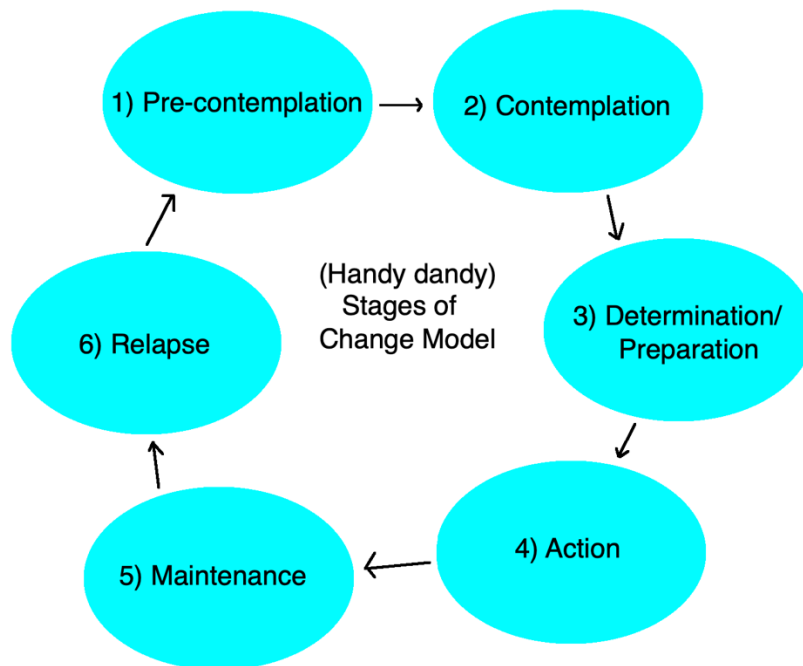
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Maintenance: The patient has stably implemented behavior to reduce certain behaviors (e.g., has been off of cigarettes for three weeks and is currently on the patch).

Relapse: The patient has experienced a recurrence (i.e., relapse) of the behavior (e.g., was on the patch for a month but is now smoking sometimes).

71. A 30-year-old woman presents to the clinic because she wants to quit smoking. She has been smoking one pack of cigarettes daily for 15 years. Recently, her mother was diagnosed with severe COPD and was prescribed continuous oxygen therapy. The patient does not wish for that to happen to her. Moreover, she recently started dating a new partner and they are thinking of having a child. She knows smoking would increase the risk of complications. She has never tried quitting before. She is asking about medications that could help her quit. Which stage of change best describes this patient?
- A) Action
 - B) Contemplation
 - C) Maintenance
 - D) Pre-contemplation
 - E) Preparation

The correct answer is E.



The patient is currently at the preparation stage. She has successfully acknowledged the risks involved with smoking and has decided to quit. Making an active decision to make the change and making a plan to do it is what differentiates the preparation and contemplation phases.

Pre-contemplation: The patient has not yet considered (i.e., contemplated) the notion of changing his or her behavior.

Contemplation: The patient has considered (i.e., contemplated) the idea of changing his or her behavior but has not yet taken any concrete or tangible steps to implement such change.

Determination/Preparation: The patient is mentally committing to change but has not yet reduced or ceased the behavior (e.g., reading pamphlets/websites about harmful effects and plans to start the nicotine patch).

Action: The patient is now taking concrete steps to reduce or stop a behavior (e.g., has commenced nicotine gum/patch). There is minimal stability to the patient's behavior at this point and chance of relapse is high.

Maintenance: The patient has stably implemented behavior to reduce certain behaviors (e.g., has been off of cigarettes for three weeks and is currently on the patch).

Relapse: The patient has experienced a recurrence (i.e., relapse) of the behavior (e.g., was on the patch for a month but is now smoking sometimes).

Medical errors and prevention

72. A 25-year-old man presents to the physician with mucopurulent discharge that first appeared several days ago. Past medical history is unremarkable. Gram-negative diplococci are seen on gram-stain. He is administered IM ceftriaxone and oral azithromycin. Within a few minutes, the patient develops a severe reaction requiring intubation. Which of the following best describes the event that happened?
- A) Medical negligence
 - B) Near-miss event
 - C) Non-preventable error
 - D) Preventable error
 - E) Swiss cheese model

The correct answer is C.

Some medical errors are not preventable. In this case, there was no reason to believe this patient would be allergic to either of these antibiotics, and the physician's treatment was appropriate in this case.

This would have been both a preventable error and negligence if the patient's allergy history had been previously documented and not followed.

The swiss cheese model is the notion that although individual measures relating to treatment or management may each carry minor risks of error (i.e., things can "fall through the holes/cracks"), when you add up a number of layers of protection and responsibility (i.e., proper communication and documentation), it becomes much less likely that an error will occur.

Near-misses are medical errors that may have occurred had events played out differently, but fortuitously did not transpire as such. Reporting of near-miss events is not required, although documentation is helpful to decrease probability of subsequent occurrences.

73. A 30-year-old athlete is about to undergo an ACL reconstruction at a major academic center. Pre-operative planning transpires uneventfully and the patient is prepared for surgery. As the surgeon is about to clean the operating field, the resident informs him that the patient is allergic to iodine. The physician then uses a different disinfectant. Which of the following best describes the event that occurred?
- A) Near-miss event
 - B) Neglect
 - C) Negligence
 - D) Non-preventable error
 - E) Preventable medical error

The correct answer is A.

This is a near-miss event because a medical error would have occurred had the resident not fortuitously intervened at the last moment. Reporting of near-misses is not required, although documentation is helpful to reduce the probability of subsequent occurrences.

Neglect is defined as failure of a designated caregiver to meet the needs of a dependent (i.e., child, elder, person of disability). This term does not relate to medical errors.

Negligence refers to gross deviations from normal treatment resulting in harm (e.g., operating on the ACL of the wrong leg).

Non-preventable errors are those that could not have been prevented despite all appropriate actions by the medical team (e.g., the patient had fibrosis around his ACL not seen on MRI, resulting in a difficult operation and the surgeon injuring the ligament further).

Preventable errors are those that can be avoided if proper documentation and communication are followed (e.g., awareness of patient allergies).

Sign-out notes

74. The on-call team at a major academic center has been notified of a 60-year-old male who is experiencing acute exacerbation of COPD. The patient's primary team has already left the hospital. In order to learn more about the patient, the on-call resident reads the sign-out and notices that drug allergies were omitted. Which of the following would decrease the risk of similar situations happening in the future?
- A) Attaching admission and progress indices to sign-out notes
 - B) Ensuring more than one resident reads each sign-out note before it is passed to the on-call team
 - C) Limiting the number of words in the sign-out note
 - D) Making sure sign-out notes are delivered in person
 - E) Using a standardized template for sign-out notes

The correct answer is E.

Standardization would most likely have prevented the omission of allergies from the sign-out note. A standardized sign-out note would certainly contain the allergies section and the primary team would not have forgotten to mention them. Checklists and standardization significantly reduce the risk of human error in most settings (e.g., a checklist before surgery).

The Agency for Healthcare Research and Quality suggests three principles of safe design:

- Standardization when possible
- Creation of independent checks
- Learning from deficits in prior designs

Making sure sign-out notes are delivered in person might help since it would enable the on-call residents to ask additional questions, however this would not be as effective as standardizing the forms.

Limiting the number of words might increase the likelihood that crucial information is omitted. Moreover, it is impossible to know how many words would be necessary to properly describe any particular patient's condition.

Attaching admission and progress indices would help but would likely be too detailed and time-consuming for practical implementation.

Prescription and medication errors

75. A 55-year-old man has been hospitalized in the ICU with sepsis. He suffers from multiple comorbidities for which he is taking a number of different medications. His hospital course is complicated by two severe drug interactions. Ultimately, the patient recovers and is discharged from hospital. Which of the following could decrease the risk of similar events happening in the future?
- A) A nurse whose job is to check every drug that has been prescribed
 - B) A pharmacist participating in rounds
 - C) An improved system of reporting interactions
 - D) Providing additional pharmacology classes to intensivists
 - E) Providing physicians with computer databases of drug interactions

The correct answer is B.

The Agency for Healthcare Research and Quality reports that a pharmacist present on rounds may lead to a decreased incidence of prescription errors.

Additional measures that lead to decreased incidence of adverse ICU outcomes are having no more than two patients assigned to a single nurse and ensuring that an intensivist is present on daily rounds.

Online databases of drug interactions already exist and can be very useful in clinical practice, but having a pharmacist available at all times reduces the incidence of prescription errors more than such databases.

76. A 75-year-old man presents to the GP for a 3-month history of tremor of his left hand. He has just moved to a new town and this is his first appointment at the clinic. His only medication is haloperidol, but the patient is unsure why it was prescribed. He has not brought any medical records with him. Which of the following is the appropriate next step by the physician?
- A) Conduct a psychiatric history
 - B) Contact the previous physician regarding the haloperidol
 - C) Discontinue haloperidol
 - D) Discontinue haloperidol and commence aripiprazole
 - E) Refer the patient back to his previous physician for care

The correct answer is A.

As part of any appropriate history and examination, this patient should also have a psychiatric assessment performed while he is in-office.

After conducting a thorough history and examination, the physician should contact the previous physician regarding the haloperidol.

Haloperidol can cause extra-pyramidal side-effects, including acute dystonia, akathisia, Parkinsonism, and tardive dyskinesia. Stopping a typical and switching to an atypical (e.g., aripiprazole) is the management for tardive dyskinesia.

Root-cause analysis

77. A 75-year-old man is being treated for a recently sustained hip fracture. Due to unbearable post-surgical pain, he was prescribed morphine. A few hours after the medication was started, a nurse notices that the patient's breathing has slowed and that he has pinpoint pupils. The patient improves after a dose of naloxone. It is later discovered that he accidentally received double the dosage of morphine. The hospital decides to perform root-cause analysis of the situation and they start by gathering data on the event. Which of the following represents the next best step in performing this analysis?
- A) Create a causal factor flowchart
 - B) Identify root causes
 - C) Implement changes
 - D) Generate recommendations
 - E) Measure success of changes implemented

The correct answer is A.

Root-cause analysis is the process of upstream/initial (i.e., root) causes of problems in order to identify appropriate solutions.

Root-cause analysis adheres to the assumption that it is more effective to systematically prevent and solve for underlying problems than it is to just address ad hoc symptoms after the fact.

Root-cause analysis consists of 4 steps:

- Collecting data (e.g., how dosages of opioids are administered)
- Creating causal factor flowcharts (i.e., which events led to the next)
- Generating and implementing recommendations (e.g., colored syringes with dosages)
- Measuring success of changes implemented (e.g., # of errors over time)

The hospital has already gathered data on the event, so the next step is to create a causal flowchart. Once the flowchart is created and it becomes evident which sequence of events led to the error, it is appropriate to develop and implement a plan to prevent similar errors from happening in the future.

Health insurance

78. A 67-year-old man presents to the office for his annual follow-up visit. His only complaint is that his blood pressure has been poorly controlled despite taking amlodipine regularly. The patient does not drink or smoke. The physician decides to prescribe lisinopril to better manage the patient's blood pressure. Which of the following insurance plans would cover the patient's prescription cost?
- A) Medicaid
 - B) Medicare Advantage
 - C) Medicare Part A
 - D) Medicare Part B
 - E) Medicare Part C
 - F) Medicare Part D

The correct answer is F.

Medicare Part D helps cover the cost of prescription drugs and some vaccines.

Medicare Part A (hospital insurance) covers inpatient services, including hospital stays, care in skilled nursing facilities, hospice care, and some home healthcare.

Medicare Part B (medical insurance) covers doctor's services, outpatient care, medical supplies, and preventive services.

Medicare Part C covers what Part A and Part B cover. Most Medicare Part C plans also cover prescription drugs, dental, vision, and hearing expenses.

Medicare Advantage Plans are "all-in-one" alternatives to Medicare, where insurance providers may offer additional services but have to follow Medicare's coverage rules.

Medicaid is a federal and state program that helps with medical costs for low-income patients.

79. A 45-year-old woman presents to her physician for a follow-up appointment after an elevated blood pressure reading two weeks ago. Her blood pressure today is 150/100 and a decision is made to commence her on amlodipine. Upon calling her insurance provider, the patient is informed that the insurance will only begin coverage of costs once she has paid \$1,000 out-of-pocket. Which of the following best describes this out-of-pocket expense referred to by the insurance company?
- A) Coinsurance
 - B) Copayment
 - C) Deductible
 - D) Out-of-pocket maximum
 - E) Premium

The correct answer is C.

A deductible is the amount of money a patient needs to pay each year before his or her insurance policy starts paying. For example, this patient's \$1,000 deductible means that she is required to cover the first \$1,000 dollars of her medical bills before her insurance provider will cover anything.

Coinsurance is the amount the patient pays to share the cost of the covered treatments after the deductible is met. For example, if the treatment's total cost is \$2,000 and there is a 20% coinsurance, the patient will need to pay \$1,200 – i.e., the \$1,000 deductible + 20% of the subsequent insurance payment (\$200 is 20% of the second \$1,000).

A copayment is a flat fee for some expenses (e.g., \$20 for every doctor visit).

A premium is the monthly fee of the health insurance policy.

An out-of-pocket maximum is the most a patient would have to pay for the services covered by a plan over the course of a year.

Breaching vs maintaining confidentiality

80. A 40-year-old woman presents to the physician for a routine health maintenance examination. She begins discussing problems she has been having in her relationship with her husband. She says she has had thoughts about wanting to have him killed. The physician probes further, and over the course of ten minutes the patient reveals that she is not joking. She knows they are not good thoughts she is having, but she keeps thinking about this idea and might hire someone. Upon recognizing the seriousness of the physician's demeanor, the patient says, "You know I'm joking right. Don't tell me for a second you thought I was serious." What is the most appropriate course of action by the physician?
- A) Communicate to the patient you will be contacting law enforcement
 - B) Communicate to the patient you will be contacting law enforcement and her husband
 - C) Communicate to the patient you will be contacting her husband
 - D) Communicate to the patient you understand relationships are difficult and will not break confidentiality
 - E) Communicate to the patient you understand relationships are difficult and then contact law enforcement

The correct answer is C.

If a physician believes a patient has sincerely conveyed intent to cause serious harm to a third party, the physician must break confidentiality and notify the third party. This most saliently applies (although is not limited to) homicidality and rape.

The “duty to warn” third parties famously stems from the 1985 *Tarasoff* case, where a patient had confided to a psychotherapist murderous intent, confidentiality was maintained, and then the patient went on to commit murder. California law now encodes the *Tarasoff* rule, in which psychotherapists have a legal duty to warn potential victims of serious harm. States differ as to their exact laws, but the duty to warn is a principle ubiquitously upheld in the medical profession since the *Tarasoff* case.

Law enforcement does not need to be contacted when mere sentiments/plans have been conveyed, since no crime has been committed yet. If a patient has confided that he or she committed serious crime in the past (e.g., murder, armed robbery), law enforcement must be contacted.

Inpatient vs outpatient management

81. A 30-year-old woman undergoing outpatient chemotherapy for Hodgkin lymphoma presents to the physician with painful vesicular lesions on her back. The physician believes that a one-day inpatient stay with IV acyclovir is most appropriate, followed by discharging the patient on oral valacyclovir. He is aware the patient has history of hospitalization and abhors being in hospital. Which of the following is the most appropriate way to initiate discussion regarding this patient's management?
- A) "Here is oral valacyclovir for your condition. This way you don't need to be in hospital."
 - B) "How do you feel about entering hospital?"
 - C) "How do you feel about receiving IV treatment in outpatient emergency for the next day?"
 - D) "I'd like to admit you to hospital, but I know you've hated that type of thing in the past."
 - E) "We're going to admit you to hospital for a short stay. Is that okay with you?"

The correct answer is B.

A simple open-ended question assessing the patient's feelings about entering hospital is most appropriate. This exact same type of statement is also used if needing to involuntarily hospitalize a patient for, e.g., suicidality.

IV acyclovir is the answer on USMLE for herpes zoster if the patient is immunocompromised, undergoing chemotherapy, or is toxic (i.e., high fever, chills). If the patient merely has diabetes mellitus, orals can be given without the need for admission to hospital (on NBME).

Choice A is wrong because this would be violating non-maleficence – i.e., we would be doing harm to the patient if not at least offering the management we best see fit.

Choice C is wrong because this represents mismanagement of where the patient needs to be treated. It might seem that this patient could be harmlessly managed in outpatient emergency, but if a true crisis were to arise and maximal staff resources were needed elsewhere, that would be doing a disservice to this patient. Patients should be accommodated in every way possible, but this does not mean improper triage or location of care.

Choice D is a good, direct statement. But it is not as ideal as B because it is not an open-ended question. Choice D might be appropriate if the patient responds to B by saying she does not want to be in hospital. Now the physician needs to directly communicate his thoughts about admitting her.

Choice E suggests an involuntary hospitalization is ensuing. This would be an acceptable answer after choice B if the patient needed to be admitted for, e.g., suicidality.

Alternative medicine

82. A 35-year-old woman was prescribed fluoxetine six weeks ago for major depression. At today's follow-up appointment she tells the physician she started taking an herbal supplement a couple days ago, as suggested by her friend, although she forgets the name of it. Which of the following would be the most appropriate response by the physician?
- A) "As your physician I have to tell you that these supplements are insufficiently regulated."
 - B) "Herbal supplements are, unfortunately, ineffective against major depression."
 - C) "How have you been feeling since starting the supplement?"
 - D) "I want you to find out for me as soon as possible what kind of supplement it is."
 - E) "Let's see how we can be work the herbal supplement in with our other treatments."

The correct answer is D.

This is a “hard” question.

The combination of St John Wort and SSRIs can lead to serotonin syndrome. It is urgent to know what kind of supplement the patient has started so that an adverse outcome can be prevented. Patients who commence anti-depressants will certainly be counseled beforehand about the dangers of taking concurrent herbal supplements, however this patient just began taking an unknown supplement and this needs to be seriously and firmly addressed.

The reason this question is hard is because choice E is often a correct answer when patients begin alternative or cultural remedies in non-urgent settings. For instance, if an immigrant patient is not taking an SSRI and wants to add “elk bone marrow powder” to her daily routine, this should not be outright rejected, but instead attempts should be made to facilitate the incorporation of cultural remedies into the patient’s care.

Choice C is wrong because the patient could feel perfectly fine right now but then develop an adverse reaction later. This does not directly address the underlying issue that the herbal supplement needs to be identified.

83. A 6-month-old boy is brought to the physician by his mother. They are recent emigrants from Sub-Saharan Africa. The mother is convinced that the child has been possessed by an evil spirit and has been keeping a bottle of water on his head to make the spirit go away. The child is diagnosed with acute otitis media and the physician would like to give oral amoxicillin. The mother is still convinced it is the spirit. Which of the following is the most appropriate way to address the situation with the patient?
- A) "Can you tell me more about this spirit? I want to understand your thoughts here."
 - B) "Do you have a psychiatric history by any chance?"
 - C) "I understand you think there is a spirit. That is okay."
 - D) "Let's add an antibiotic to what you are already doing to help your child."
 - E) "Sometimes we might attribute illness to things that really are not extant."

The correct answer is D.

This child needs oral amoxicillin irrespective of the mother's belief about the spirit.

It must be understood that patients of different cultural and religious backgrounds will sometimes carry beliefs not shared by the physician. Patients and their families should not have their views on alternative remedies outright rejected. If the remedies are otherwise harmless, the physician should discuss how the remedy and necessary treatment can be worked in together. (This is similar to Choice E of question 82)

If the patient wishes to incorporate a remedy the physician knows little about, he or she can do research on the remedy (i.e., literally searching on Google) while the patient is there in order to develop a plan. This is because the aim is to be sensitive, and not immediately dismissive of, cultural beliefs.

Choice C sounds reasonable and is appropriate as an isolated statement. But D is superior because it implies an acceptance of the mother's beliefs (i.e., it doesn't deny or refute them) while also addressing the important management of the child.

Choice B is insensitive and communicates poor understanding of cultural beliefs.

Choice E is patronizing and verbose.

Refusal of vaccination

84. A 4-month-old boy is brought to the pediatrician by his parents. The physician recommends a number of vaccinations but the parents decline and are firm believers that vaccines cause autism. "We don't want to discuss this further," they say. Which of the following is the most appropriate statement by the physician?
- A) "I am going to document this in the chart. If you change your mind, vaccines will be available."
 - B) "I am going to report this to child protective services."
 - C) "No problem."
 - D) "There is no evidence to suggest vaccines cause autism."
 - E) "We don't have to discuss this further. But you should know that vaccines do not cause autism."

The correct answer is A.

The parents have firmly stated that they do not wish to discuss this further. They have their beliefs and these must be respected. Forcing unwanted communication in this setting can damage the doctor-patient relationship. Parents have the right to refuse vaccines for their kids, even though population health data is overwhelmingly conclusive as to the benefits. In addition, vaccines do not cause autism.

The parents' refusal should be documented. This documentation in and of itself does not need to be communicated to the parents (choice A), but the reason this is correct over choice C is because the parents are made aware that if they change their mind, they can still access appropriate care for their children without an issue.

Choice C carries outright resignation.

Choices D and E can come across argumentative in this setting.

Communication with patients in cognitive decline

85. A 76-year-old man comes to the office accompanied by his wife for a follow-up appointment. He was recently diagnosed with Alzheimer disease and was commenced on donepezil and memantine. The patient was in the army for many years and until recently led a very independent life. His wife says that he tried making coffee this morning and that, when she tried to help him, he got angry at her. The physician believes that maintaining written notes and labeling items around the house with colored text might help. The patient says, "You think I'm stupid don't you. I'm perfectly independent and my own man. When I was in the army guys like you took orders from guys like me." Which of the following is the most appropriate response by the physician?
- A) "Do you feel like at night when the sun goes down things get harder at home?"
 - B) "How do you feel things are going on your new medications?"
 - C) "I want to discuss with you ways I think we can make stuff at home easier."
 - D) "I'm going to discuss with your wife about keeping ambient lights on at night."
 - E) "We as guys need to make sure we keep our repertoire of females handled."

The correct answer is C.

The physician should simply communicate to the patient that he or she is attempting to help make things easier at home. The physician should be sensitive to patients in cognitive decline who are sometimes aware of their evanescent independence.

Color-coding and using labels may help patients with maintaining functionality at home for as long as possible.

Sundowning is worsening of dementia at night, where the patient can develop delirium-like activity. Ambient noise and distractions should be reduced, but use of night lights or keeping rooms brightly illuminated is incorrect (2CK Psych NBME).

Sleep hygiene

86. A 23-year-old woman comes to the physician because of an 8-week history of emotional lability, irritability, and fatigue. She says that she is not able to fall asleep readily despite feeling sleep-deprived. She is a law student and has been busy studying throughout the semester. Physical examination is unremarkable. Which of the following is the most appropriate recommendation by the physician?
- A) "Don't get out of bed when you're having trouble falling asleep."
 - B) "Exercising more can help you fall asleep."
 - C) "Sleep late whenever you can in order to catch up on sleep."
 - D) "Sometimes having a glass of wine before bed can be helpful."
 - E) "Try getting up at the same time every day, even if you feel tired."

The correct answer is E.

In order to establish healthy sleep hygiene, patients should be told to try to get up at the same time every day, even if it is difficult to do so and they feel tired.

The result is they may be more inclined to go to bed earlier in the evening, which in turn will make getting up at the same time easier, since they will have gotten more sleep per night.

People's sleep cycles respond differently to exercise at particular times of the day. Even if the influence of exercise is considered as a possibility in this patient's sleep hygiene, she should still first be encouraged to get up at the same time every day.

Patients should not be encouraged to develop an alcohol dependence in order to fall asleep.

Sensitive examinations

87. A 34-year-old woman at 34 weeks' gestation presents to emergency with copious vaginal bleeding and severe pelvic pain. She has not noticed any fetal movements during the past 24 hours. Blood pressure is 95/55. Heart rate is 110/minute. Her obstetrician is a woman who lives an hour away. The patient is dressed in a burka and explains that she is religious and does not want to be examined by a male physician. The only physician available is a man. Which of the following is the most appropriate action by the physician?
- A) Ask a female nurse to perform the examination and report the findings to the physician
 - B) Ask the patient if she would permit the examination if her husband is present at all times
 - C) Do not perform the examination and implement conservative management
 - D) Obtain emergency permission from the hospital judicial committee
 - E) Wait until the patient's obstetrician arrives

The correct answer is B.

The examination needs to be performed by a physician. The next best step is to ask the patient if she would allow the examination if her husband is present at all times. If she declines, her wishes must be respected as per autonomy.

Choice C is wrong because this is clearly an emergency situation. The patient has the right to decline an examination, but before conservative management is implemented, she should be asked if she would allow the examination with her husband present at all times.

Choice E is not practical in an emergency scenario.

Unexpected results in relation to demographic

88. An 81-year-old woman is brought to the physician by her daughter to discuss the results of a cognitive work-up. The patient has had worsening anterograde amnesia for the past 6 months. Her Mini-Mental State Exam (MMSE) score is 22/30. Serum studies demonstrate a positive result for fluorescent treponemal antibody. Which of the following is the most appropriate action by the physician?
- A) Discuss the results with the daughter only
 - B) Discuss the results with the daughter and patient together
 - C) Discuss the results with the patient only
 - D) Obtain serum TSH
 - E) Repeat testing and follow-up in 3 days

The correct answer is C.

The patient has a positive test for syphilis, suggesting that her cognitive dysfunction might be due to neurosyphilis. These results should first be discussed with the patient only. Her decision-making capacity should be assessed during the examination and documented accordingly.

Just because the patient has a low MMSE score does not in and of itself warrant that the daughter be present during the disclosure of test results. The patient's right to confidentiality must be respected.

If the patient ultimately refuses treatment for syphilis, this must be respected and also documented.

Hypothyroidism and B12 deficiency are reversible causes of dementia, however this patient is positive on syphilis screening, so she should be treated with penicillin as the next best step in management.

Breast masses in teenagers

89. A 13-year-old girl is brought to the physician by her mother because of a 3-cm rubbery mass palpable beneath the left nipple. She is Tanner stage 3. Family history is remarkable for the patient's grandmother passing away from breast cancer at age 57. Which of the following is the most appropriate communication by the physician?
- A) "How do you feel about having a biopsy of the mass?"
 - B) "How do you feel about having your blood drawn so we can check some hormone levels?"
 - C) "I know you must feel really concerned about this. A biopsy is most appropriate."
 - D) "This typically resolves over 12-18 months."
 - E) "What is your understanding of breast masses?"

The correct answer is D.

Breast masses are normal findings *in both boys and girls* during puberty. This is referred to as physiologic gynecomastia and typically resolves within 12-18 months.

Sometimes one breast might develop a lot more quickly than the other. This is also benign and referred to as virginal breast hypertrophy.

Hormones do not need to be checked in this girl as she is otherwise developing fine. It is high-yield to know that once a girl reaches Tanner stage 3, menarche is imminent (this is a direct answer on a 2CK obgyn NBME). The USMLE will frequently trap students into choosing answers relating to “check hormones,” when in reality observation is frequently the answer in pediatrics questions.

Medical students disclosing information

90. A medical student is caring for a 68-year-old man who has been admitted to hospital for biopsy of a pancreatic mass. The resident physician asks the student not to discuss results with the patient until the oncologist has evaluated the findings. During preparation for ward rounds, the student goes to the patient's bedside to grab the drug chart and notices the electronic medical chart was updated by Pathology ten minutes ago. The biopsy result shows adenocarcinoma. The patient is awake and asks the student, "What did my biopsy show?" Which of the following is the most appropriate response by the student?"
- A) "I'm just a student, sir."
 - B) "I'm not sure, but the oncologist is seeing you today."
 - C) "I'm not sure. Let me grab the resident physician."
 - D) "Let me grab the resident physician so we can discuss the results."
 - E) "What is your understanding of the biopsy process?"

The correct answer is D.

The student was given instructions by a senior to not disclose the results. Professional conduct must be upheld and the results not disclosed.

Choice A ignores the patient's question. Even if the implication is, "I'm just a student, so a doctor should discuss the results with you," this in turn should be communicated, as with Choice D.

Choices B and C are wrong because the student is lying to the patient. This is never acceptable.

Choice E is a nominal statement that might be part of the process of breaking bad news, but in this case the student should simply get the resident physician in order to discuss the results.



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